

Operationalising the Three-Axes Framework: A Composite-Indicator Companion to *AI-Proliferation and Middle Powers*

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Companion to the working paper *AI-Proliferation and Middle Powers: Preparation and Response Mechanisms* (2026)

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Abstract

The working paper *AI-Proliferation and Middle Powers* argues that fourteen named middle powers differ along three axes—*capacity depth*, *governance orientation*, and *infrastructure posture*—and that the tractability of preparedness actions depends on a country’s position along them, yet it never empirically positions any country. This companion study closes that gap with the Middle-Power AI-Proliferation Preparedness Index (M-PAPI): a reproducible composite indicator built from fourteen publicly sourced indicators following the OECD/JRC *Handbook on Constructing Composite Indicators*. We report a literature-weighted ranking in which the United Kingdom (2.214) and South Korea (2.162) co-lead—the UK first under all three cardinal schemes (equal, PCA, and literature), with South Korea a close second, while the weighting-free Borda/Kemeny consensus instead places Korea first—and India (0.104) and Taiwan (0.060) trail; the top-five set {UK, Korea, France, EU, Japan} is identical across equal-, PCA-, and literature-weighting schemes. Six hypotheses are pre-registered with numeric pass criteria and resolved mechanically: two are fully supported (cluster heterogeneity; insensitivity of the top tier to any single analyst-coded indicator) and four are partially supported with specific, named caveats (cross-axis separability, axis dimensionality, ranking robustness, and tier-membership stability). A vulnerability overlay maps axis scores onto the paper’s three attack vectors, counterfactual scenarios quantify per-country policy leverage, and an enabler-readiness matrix extends the paper’s §4 preparedness checklist into a structural readiness score for every country. The ranking is strongly rank-concordant with the independent Oxford Insights Government AI Readiness Index 2024 (Spearman $\rho = 0.852$ over the thirteen jurisdictions both cover), evidence of external concurrent validity. Within a pinned software environment, all outputs regenerate byte-for-byte under a fixed seed and an offline data snapshot. The index empirically grounds the paper’s §1–§2 premise—the middle powers are differentiable, and the most-exposed tier is identifiable—and structurally operationalises the paper’s §4 tractability claim, while explicitly declining the §3 incident-level claims it cannot yet test.

Keywords: composite indicators; AI governance; middle powers; AI proliferation; reproducible research; sensitivity analysis.

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1 Introduction

The development of frontier artificial intelligence is concentrated in a small number of laboratories in the United States and China, leaving *middle powers*—states with meaningful but non-dominant international influence—largely in the position of responding to, rather than shaping, the technology’s trajectory. The working paper *AI-Proliferation and Middle Powers: Preparation and Response Mechanisms* (Teague et al., 2026) argues that, as access to highly capable model weights diffuses, middle powers face a structurally under-addressed vulnerability to three proliferation-enabled attack vectors—cyber-attacks, chemical/biological/radiological/nuclear (CBRN) threats, and influence operations—and that national-level preparedness is a more tractable priority than contesting the frontier. To organise that argument the paper distils two widely cited “building blocks of AI power” frameworks into three axes along which middle powers differ: *capacity depth*, *governance orientation*, and *infrastructure posture*.

The paper names fourteen middle powers—the United Kingdom, Canada, France, Germany, Japan, India, Israel, Singapore, South Korea, Sweden, Saudi Arabia, the United Arab Emirates, the European Union, and Taiwan (Wang et al., 2024)—and repeatedly invokes “tractability depending on a country’s position within the three axes,” but it never *positions* any country on those axes. This companion study supplies that missing empirical layer.

Contribution. We construct the Middle-Power AI-Proliferation Preparedness Index (M-PAPI), a transparent composite indicator that places each of the fourteen middle powers on the three axes using fourteen publicly available indicators, and we report per-country rankings, a per-vector vulnerability overlay, and counterfactual policy scenarios. The study is deliberately *not* new theory; it is empirical scaffolding that makes the working paper’s per-country claims auditable, replicable, and updatable. We follow the OECD/JRC *Handbook on Constructing Composite Indicators* (OECD and European Commission, Joint Research Centre, 2008) throughout and report three weighting schemes side by side with a full sensitivity battery.

Scope. M-PAPI operationalises the paper’s §1 (axes and country set) and §2 (attack vectors) directly, and the §4 *tractability* claim structurally. The paper’s §3 (trigger-event ladder) is *not* operationalised: a country-year, AI-attributed incident dataset would be required, and none exists for these fourteen jurisdictions across the three attack vectors without heuristic estimation (Section 7). The §4 detection/escalation/mitigation checklist is operationalised *partially*: its actions appear both as the counterfactual policy scenarios of Section 5 and, more generally, as the enabler-readiness matrix of Section 5.7, which scores each country’s structural readiness for the checklist actions

Table 1: The three axes of the working paper’s framework and the M-PAPI indicators assigned to each. The working paper names the third axis *Compute and Infrastructure Posture*; the headline uses the ICT measures listed, and a dedicated TOP500 compute indicator is sourced and tested as a robustness check (Appendix B). Indicator construction and sources are detailed in Section 3.

Axis	Definition (working paper §1)	Indicators
Capacity Depth	Domestic technical talent, AISI-equivalent institutions, AI R&D output	C1 notable models; C2 training compute; C3 AI publications; C4 AI patents; C5 AISI presence
Governance Orientation	AI-governance maturity, alliance posture, bilateral lab agreements	G1 national AI strategy; G2 ITU GCI; G3 V-Dem liberal democracy; G5 IGSC member firms; G6 Australia Group
Infrastructure Posture	Domestic compute, ICT infrastructure, platform/cloud presence	I1 ITU IDI; I2 broadband; I3 secure servers; I5 ND-GAIN readiness

against the indicators proxying each action’s required enablers—not as country-year preparedness outcomes. The index measures *relative* position within the fourteen-country reference frame; it does not claim that a high score means a country is “safe.”

Pre-registration. Six falsifiable hypotheses (H1–H6) are stated with numeric pass criteria before estimation (Table 6) and resolved mechanically against those criteria. Reporting the partial-support outcomes explicitly—rather than only the favourable ones—is a deliberate feature of the design.

2 Conceptual framework

2.1 The three axes

Table 1 reproduces the paper’s axis definitions and lists the indicators assigned to each. The fourteen indicators comprise five *capacity-depth* measures, five *governance-orientation* measures, and four *infrastructure-posture* measures. The working paper labels the third axis *Compute and Infrastructure Posture*; the headline operationalises it as *Infrastructure Posture* from ICT measures, while a dedicated, hosting-attributed compute indicator (the TOP500 supercomputer-system count) is sourced and tested in Appendix B—it leaves the ranking essentially unchanged (Spearman $\rho = 0.996$) and is reported as a robustness check rather than folded into the pre-registered headline (Section 7).

2.2 The three attack vectors

The paper’s §2 argues that each attack vector loads the three axes differently. We encode this as a per-vector set of across-axis weights (Table 2); these weights are the authors’ translation of the paper’s §2.2–§2.4 qualitative arguments, not externally elicited, and their effect on the per-vector ranking is stress-tested in Section 5. Cyber threats emphasise defensive capacity and monitoring infrastructure; CBRN threats emphasise capacity and biosecurity governance; influence operations emphasise open distribution infrastructure and democratic-resilience governance.

Table 2: Per-vector across-axis weights (each row sums to 1), translating the working paper’s §2.2–§2.4 arguments. Used only for the vulnerability overlay (Section 5); the headline composite does not depend on them.

Vector	Capacity	Governance	Infrastructure
Cyber (§2.2)	0.40	0.25	0.35
CBRN (§2.3)	0.50	0.40	0.10
Influence operations (§2.4)	0.20	0.35	0.45

2.3 Why a composite, and its limits

Composite indicators are routinely criticised for their dependence on weighting. Following [OECD and European Commission, Joint Research Centre \(2008\)](#), our defence is not to claim the weights are uniquely correct, but to (i) report three weighting schemes side by side, (ii) quantify rank stability under Monte Carlo perturbation, (iii) make the indicator-to-axis mapping fully transparent, and (iv) maintain end-to-end source traceability. The handbook is explicit that a composite should never be presented as a single number without accompanying sensitivity ranges; we adhere to this throughout.

3 Data

3.1 Indicator set and provenance

Table 3 lists the fourteen indicators with their source, pre-normalisation transform, and within-axis literature weight. Twelve of the fourteen are fully sourced from Tier-1 institutional providers with no analyst-coding step (eleven retrieved programmatically; one—Australia Group membership, G6—inline against its cited public roster ([Australia Group, 2024](#)) because the upstream page is unreliable for automated fetch). Programmatic sources include Epoch AI ([Epoch AI, 2024](#)) for notable-model counts and training compute (C1, C2), OpenAlex ([Priem et al., 2022](#)) for AI publications (C3), the Stanford AI Index ([Maslej et al., 2025](#)) for AI patents per 100,000 inhabitants (C4), the ITU Global Cybersecurity Index ([International Telecommunication Union, 2024a](#)) (G2), V-Dem ([Coppedge et al., 2026](#)) for the liberal-democracy index (G3), the IGSC roster ([International Gene Synthesis Consortium, 2026](#)) (G5), the ITU ICT Development Index ([International Telecommunication Union, 2024b](#)) (I1), the World Bank WDI ([World Bank, 2024](#)) for broadband and secure servers (I2, I3), and the ND-GAIN Country Index ([University of Notre Dame, 2026](#)) (I5).

Two indicators carry an analyst-coding step and are flagged `requires_verification` in the project’s data registry: G1 (national AI-strategy comprehensiveness, a 0–3 rubric over the OECD.AI Policy Observatory ([OECD, 2024](#))) and the ordinal extension of C5 (AIS presence; the binary 0/3 split between Network and non-Network members is authoritative from the NIST fact sheet ([U.S. Department of Commerce, 2024](#)), while the intermediate codes for non-members are authorial). For each, per-country coding protocols and evidence citations are recorded in the notebook, and Section 5 shows that collapsing each indicator to its authoritative core leaves the ranking unchanged. A fifteenth candidate indicator—a bilateral frontier-lab MoU count (G4)—was excluded from the scored composite for lack of a reproducible authoritative source: there is no public registry of such MoUs, the landscape blurs MoU, partnership, and commercial deal, and documented cases often involve a national AI Safety Institute, double-counting C5. The bilateral-lab-MoU concept is

Table 3: The fourteen indicators: axis assignment, source, pre-normalisation transform, and within-axis weight under the literature scheme. Weights are specified as relative weights and renormalised to sum to one within each axis before aggregation (Eq. 2); the governance weights shown sum to 0.90 as specified and are rescaled accordingly. Polarity is “higher is better” throughout.

ID	Indicator (source)	Axis	Transform	Lit. wt.
C1	Notable models, 5-yr (Epoch AI)	Capacity	log1p	0.30
C2	Training compute, 5-yr (Epoch AI)	Capacity	log1p	0.25
C3	AI publications (OpenAlex)	Capacity	log1p	0.10
C4	AI patents per 100k (Stanford AI Index)	Capacity	log1p	0.10
C5	AISI presence (NIST fact sheet)	Capacity	none	0.25
G1	National AI strategy (OECD.AI; coded)	Governance	none	0.20
G2	Global Cybersecurity Index (ITU)	Governance	none	0.30
G3	Liberal-democracy index (V-Dem)	Governance	none	0.20
G5	IGSC member firms (IGSC roster)	Governance	none	0.10
G6	Australia Group membership (roster)	Governance	none	0.10
I1	ICT Development Index (ITU IDI)	Infrastructure	none	0.40
I2	Fixed broadband per 100 (World Bank)	Infrastructure	none	0.20
I3	Secure servers per million (World Bank)	Infrastructure	log1p	0.20
I5	ND-GAIN readiness (Notre Dame)	Infrastructure	none	0.20

retained qualitatively; excluding G4 leaves the United Kingdom leading, with South Korea a close second.

3.2 Missingness and observability

The raw 14×14 matrix is 93.9% observed (184 of 196 cells). Figure 1 maps every cell to one of three states: observed, stage-1 axis-mean imputed, or stage-2 column-mean fallback. The largest gap is C4 (AI patents), missing for six of the fourteen jurisdictions because they fall below the Stanford AI Index’s published top-15 cut-off; C4 therefore exceeds the imputation threshold and its missing cells are column-mean filled. Taiwan’s four infrastructure indicators are unobserved (a missing-not-at-random exclusion from ITU and World Bank ICT collection) and are likewise column-mean filled—a caveat carried through every downstream figure.

4 Methodology

The pipeline follows the OECD/JRC ten-step process (OECD and European Commission, Joint Research Centre, 2008): theoretical framework (Section 2), data selection (Section 3), imputation, multivariate analysis, normalisation, weighting, aggregation, robustness, back-to-the-data interpretation, and visualisation.

4.1 Cleaning, transforms, and imputation

Count-like indicators (C1–C4, I3) receive a $\log(1+x)$ transform; bounded scores and ordinals are left untransformed. The European Union row is constructed from member states—summed for count indicators that upstream sources report per country, and member-mean for bounded scores and

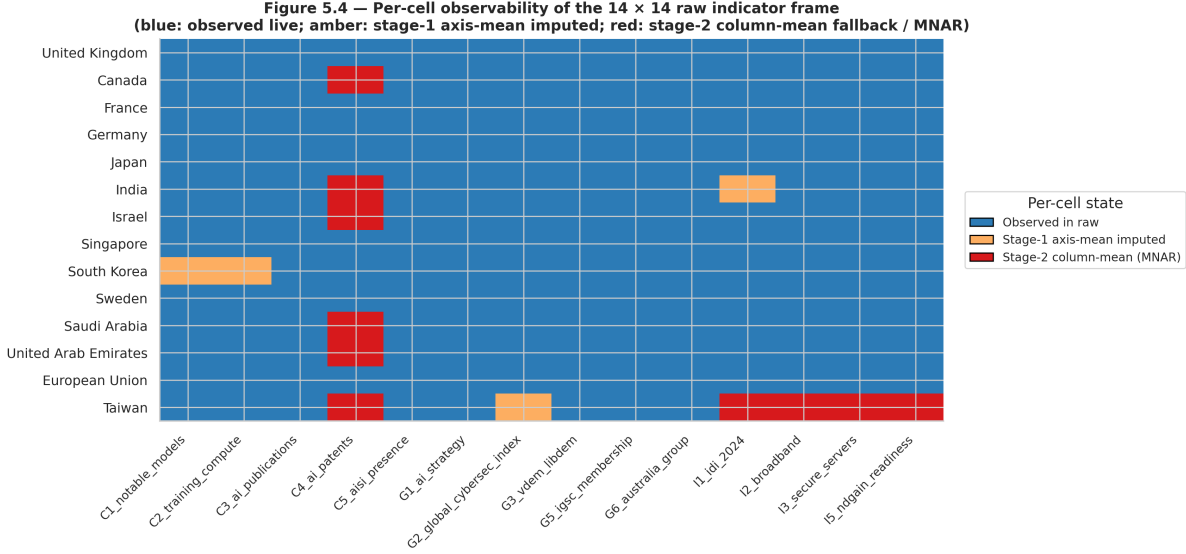


Figure 1: Per-cell observability of the 14 × 14 raw indicator matrix. Blue: observed; amber: stage-1 axis-mean imputed; red: stage-2 column-mean fallback (missing-not-at-random). C4 patents (six jurisdictions) and Taiwan’s four infrastructure cells dominate the red region. Source: `outputs/missingness_audit.csv`.

per-capita rates—a deliberate but internally asymmetric construction that we flag as a limitation (Section 7). Missing cells whose indicator has at most 30% missingness are imputed by the country’s average z -position across its other same-axis indicators (stage 1); indicators above that threshold, and cells with no observed axis-mate, fall through to column-mean imputation (stage 2). Only two cells receive stage-1 imputation.

4.2 Normalisation and aggregation

Each indicator is normalised by z -score across the fourteen jurisdictions,

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{\sigma_j}, \quad (1)$$

with min–max normalisation retained as a sensitivity check. Within each axis a , the axis score is a linear weighted sum of its normalised indicators,

$$A_{ia} = \sum_{j \in a} w_j z_{ij}, \quad \sum_{j \in a} w_j = 1. \quad (2)$$

Across axes, the composite is a weighted *geometric* mean, which penalises imbalance so that no single axis fully compensates for weakness on another—consistent with the paper’s §4 logic that all three axes matter. Because z -scored axis scores can be negative, each axis is shifted to positive support before the logarithm:

$$C_i = \exp \left(\sum_a W_a \log(A_{ia} - \min_k A_{ka} + \varepsilon) \right), \quad \sum_a W_a = 1, \quad (3)$$

with $\varepsilon = 10^{-3}$. The shift makes per-country composite *values* comparable only within the fourteen-country frame; the *ranking* is the portable quantity, and the sensitivity of the bottom tier to ε is quantified in Section 5.

Figure 8.1 — Horn's parallel analysis per axis (N = 10,000 permutations)

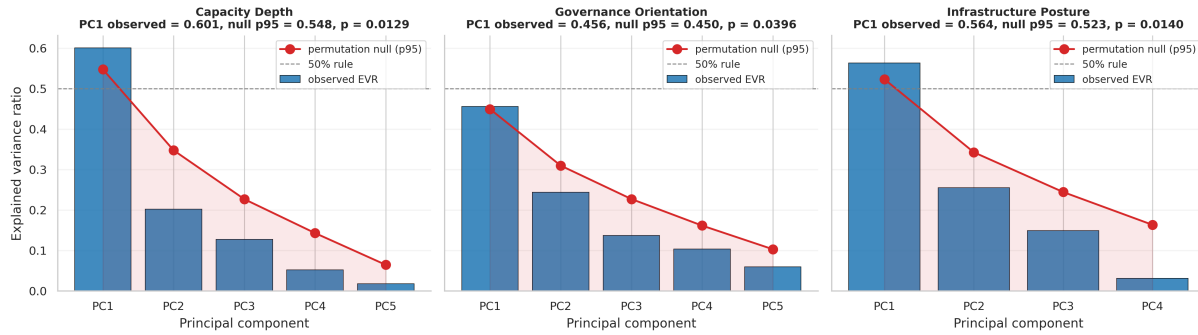


Figure 2: Horn's parallel analysis per axis (10,000 permutations). Bars: observed explained-variance ratio per component; red line: 95th-percentile permutation null; dashed grey: the 50% heuristic. The first component clears the null for governance and infrastructure; capacity is untestable (fit on $n = 8$ complete cases, below the ten-case testability floor). Source: `outputs/parallel_analysis.csv`.

4.3 Weighting schemes

Three schemes are reported side by side: **equal** (uniform within and across axes); **PCA-derived** (within-axis weights proportional to squared first-principal-component loadings, computed at run time); and **literature-informed** (Table 3; across-axis weights 0.40/0.30/0.30), round-number weights assigned by the authors to reflect the relative emphasis the GMF *Pivotal Powers* (de Hoop Schaffer et al., 2024), Chatham House sovereign-AI (Sandoval et al., 2026), and Tortoise Global AI Index (Tortoise Media, 2023) frameworks place on each indicator family — not a formal expert elicitation; their influence is bounded by the sensitivity battery (Section 4) and the weighting-free Borda/Kemeny consensus. The PCA scheme is presented as a sensitivity benchmark only: at $n = 14$ its loadings are unstable, and on the governance axis it effectively zeroes the cybersecurity indicator (G2) and reallocates its mass chiefly to G3 and G6.

4.4 Multivariate diagnostics

We assess whether each axis coheres as a one-dimensional construct using within-axis correlations, Cronbach's α (Cronbach, 1951), and Horn's parallel analysis (Horn, 1965) (Figure 2). Parallel analysis compares each axis's observed first-component explained-variance ratio against a 95th-percentile permutation null over 10,000 shuffles. The infrastructure and governance axes clear the null ($p = 0.014$ and $p = 0.040$); the capacity axis is fit on only eight complete cases—below the ten-case testability floor—so its parallel-analysis verdict is suppressed as unstable (its p swings from ≈ 0.08 at $n = 7$ to ≈ 0.01 at $n = 8$). Cronbach's α is 0.760 (capacity), 0.579 (governance, below the conventional 0.7 threshold), and 0.722 (infrastructure).

4.5 Sensitivity, robustness, and inference

Rank stability is probed with: a 10,000-draw Dirichlet($\alpha=1$) Monte Carlo over within- and across-axis weights; leave-one-indicator-out (LOO); a z -score-versus-min-max normalisation swap; a bibliographic-classifier swap (broad-AI versus narrow-ML OpenAlex concept for C3); an ε -grid; a geometric-versus-arithmetic aggregation-function swap; a C5-axis-reassignment check; and an analyst-coded judgment-collapse check. All rank correlations are reported with Bonett-Wright

(Bonett and Wright, 2000) Fisher- z 95% confidence intervals, whose standard error

$$\text{SE} = \sqrt{\frac{1 + \rho^2/2}{n - 3}} \quad (4)$$

is the variance-stabilised form appropriate for Spearman’s ρ (wider than the naive Pearson $1/\sqrt{n - 3}$). A second, *joint* Monte Carlo perturbs the imputed cells alongside the weights—stage-1 axis-mean cells by the country’s observed within-axis z -dispersion, and stage-2/MNAR cells by $N(0, \sigma_2)$ with $\sigma_2 \in \{0.5, 1.0\}$ —while holding every *observed* value fixed, yielding per-country credible rank intervals and tier-membership probabilities (Section 5, Figure 5). Concurrent validity is assessed against an independent external index (Section 6.1).

Country-influence and weighting-free checks. Two further checks target the small- n leverage and weight-dependence concerns head-on. A leave-one-*country*-out jackknife (Quenouille, 1949; Tukey, 1958) drops each of the fourteen jurisdictions in turn, re-normalises the remaining thirteen (z -scores are sample-relative), and recomputes both the ranking and the k -means typology ($k = 3$ held fixed to isolate the country-drop effect from model selection), quantifying how much any single country drives the order (Spearman ρ with Bonett–Wright CI on the retained thirteen) and the partition (adjusted Rand index against the full typology). Separately, a *weighting-free* consensus ranking treats each of the fourteen indicators as one equal ballot and aggregates the fourteen ballots with no cardinal weights—neither within- nor across-axis—via the Borda count (de Borda, 1781) and the exact Kemeny–Young median ranking (Kemeny, 1959; Young, 1988), the order minimising total Kendall- τ disagreement with the ballots, solved as a mixed-integer programme over the fourteen alternatives. If the headline survives an aggregation that removes the weighting step entirely, the ordering is a property of the data rather than of the weights.

4.6 Typology and interpretability

Country archetypes are obtained by k -means with a silhouette-selected k (Rousseeuw, 1987), cross-checked against hierarchical clustering (adjusted Rand index, Hubert and Arabie, 1985) and a multidimensional-scaling embedding (Kruskal, 1964). Per-indicator contributions to the headline composite are decomposed exactly by enumerating all 2^{14} Shapley (Shapley, 1953) coalitions, complemented by permutation feature importance and, over the same coalition set, exact second-order Shapley *interaction* indices (Grabisch and Roubens, 1999) that quantify pairwise indicator redundancy (Appendix C, Figure 25). Internal consistency, clustering, and MDS use `scikit-learn` (Pedregosa et al., 2011); the stack also includes NumPy (Harris et al., 2020), pandas (McKinney, 2010), SciPy (Virtanen et al., 2020), Matplotlib (Hunter, 2007), and seaborn (Waskom, 2021).

5 Results

5.1 Headline ranking

Table 4 reports the composite under all three schemes. Under the literature scheme the United Kingdom (2.214) and South Korea (2.162) co-lead—the UK ranks first under all three cardinal schemes (equal 2.108, PCA 2.487, literature 2.214), with South Korea a close second; the weighting-free Borda and Kemeny consensus instead place Korea first, and both carry a Monte Carlo median rank of 2, so the pair is best read as a close 1–2 pair with the leader depending on aggregation

Table 4: Headline composite scores under the three weighting schemes, ordered by the literature scheme. The top-five set is invariant across schemes. Source: `outputs/index_baseline.csv`.

Rank	Country	Equal	PCA	Literature
1	United Kingdom	2.108	2.487	2.214
2	South Korea	2.039	2.287	2.162
3	France	1.790	2.287	1.958
4	Japan	1.705	2.068	1.769
5	European Union	1.761	2.339	1.768
6	Singapore	1.346	1.668	1.767
7	Germany	1.329	1.826	1.398
8	Canada	1.357	2.008	1.350
9	Sweden	0.875	1.278	1.037
10	United Arab Emirates	0.314	0.380	0.883
11	Saudi Arabia	0.108	0.115	0.813
12	Israel	0.335	0.899	0.135
13	India	0.078	0.100	0.104
14	Taiwan	0.077	0.117	0.060

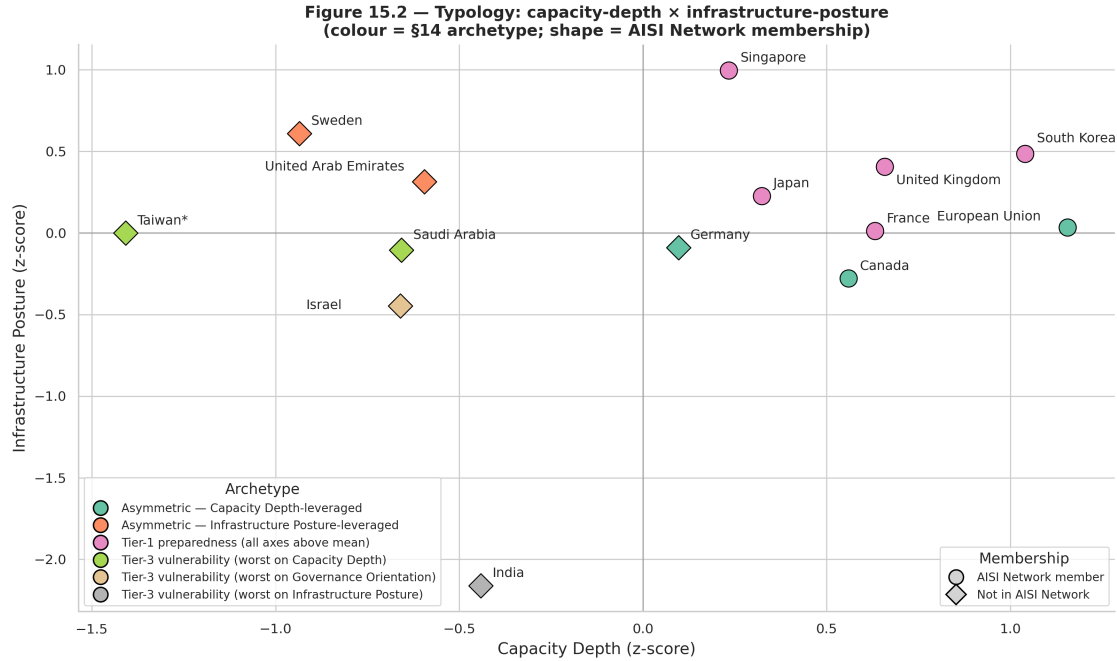
Table 5: Robustness of the headline ranking. Spearman ρ against the literature ranking with Bonett–Wright Fisher- z 95% CIs ($n = 14$). Source: `outputs/robustness_summary_with_ci.csv`.

Perturbation	ρ	Threshold	95% CI	Verdict
Weighting: equal vs PCA	0.969	0.70	[0.877, 0.993]	clears
Weighting: equal vs literature	0.969	0.70	[0.877, 0.993]	clears
Normalisation: z vs min–max	0.965	0.70	[0.861, 0.992]	clears
Classifier: AI-broad vs ML-narrow	0.960	0.85	[0.845, 0.990]	borderline

method—and India (0.104) and Taiwan (0.060) trail. The top-five set {UK, Korea, France, EU, Japan} is identical across all three schemes; the bottom-three set contains {India, Taiwan} under every scheme, with the third slot Israel under the literature scheme and Saudi Arabia under the equal and PCA schemes. Figure 3 positions the countries on the capacity $\times$ infrastructure plane, colour-coded by archetype and shape-coded by AISI Network membership.

5.2 Robustness

Under the Bonett–Wright standard error, the equal-vs-PCA, equal-vs-literature, and normalisation perturbations all clear the 0.70 robustness floor with comfortable margins, and their CI lower bounds (0.877, 0.877, and 0.861) also clear the stricter 0.85 bar (Table 5, Figure 4). Only the bibliographic-classifier check ($\rho = 0.960$, CI lower bound 0.845) sits just below the 0.85 bar imposed on the classifier (a core measurement choice rather than a downstream weight perturbation), while still clearing the 0.70 floor. The Monte Carlo rank distributions (Appendix Figure 16) show the United Kingdom and South Korea co-leading at a median rank of 2, France and the EU sharing a median rank of 4, and Japan at median 5.



* Taiwan's infrastructure-posture coordinate is column-mean-imputed (four infrastructure indicators are unobserved for Taiwan; see §14 observability caveat).

Figure 3: Country positions on Capacity Depth × Infrastructure Posture (z -scores). Colour encodes the per-country archetype derived from each country's own axis profile (Tier-1; two Asymmetric variants; three Tier-3 variants by worst axis); marker shape encodes AISI Network membership. Taiwan's infrastructure coordinate is column-mean-imputed. Source: `outputs/typology.csv`.

Rank uncertainty. The joint Monte Carlo—perturbing weights *and* imputed cells while holding observed values fixed—attaches an uncertainty band to every position. Under the most conservative setting ($\sigma_2 = 1.0$), the United Kingdom and South Korea each sit in the top five in about 98% of draws (co-leading at a median rank of 2); the bottom tier is weaker (India and Taiwan in the bottom three in 76% and 61% of draws). The added imputation noise widens the credible intervals only marginally over the weight-only Monte Carlo (Figure 5), confirming that the headline ordering is not an artefact of the fixed imputed values—the cells the pipeline had to fill do not drive the ranking. The middle of the table is where the bands overlap most, consistent with the single-slot instability behind H6's partial verdict.

Country influence and weighting-free consensus. Two further checks (Section 4) confirm the ordering is neither a single-country artefact nor a weighting artefact. The leave-one-*country*-out jackknife shows that dropping any one jurisdiction shifts the retained thirteen by at most three rank positions (worst case: dropping India or Taiwan), the retained-order Spearman ρ never falls below 0.945, and the k -means typology is identical (ARI = 1.0) for twelve of the fourteen drops—dipping only when India (ARI = 0.61) or Taiwan (ARI = 0.47) is removed. The weighting-free consensus is equally reassuring: aggregating the fourteen indicator ballots with *no* cardinal weights reproduces the literature ordering under both the Borda count (Spearman $\rho = 0.977$, 95% CI [0.906, 0.994]) and the exact Kemeny–Young median ranking ($\rho = 0.965$, [0.861, 0.991]). South Korea is first under both and the top-five and bottom-three sets are preserved under Borda; the only consequential disagreement is that the Kemeny median elevates the EU into the second slot (literature fifth)—a

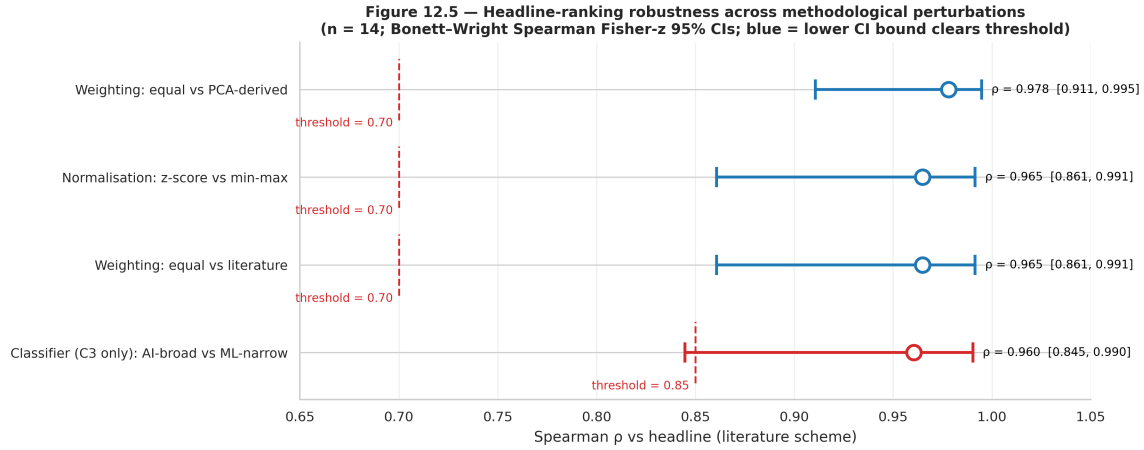


Figure 4: Robustness forest plot: Spearman ρ point estimates with Bonett–Wright Fisher- z 95% CIs against each perturbation’s pass threshold (dashed). The equal-vs-PCA, equal-vs-literature, and normalisation checks clear 0.70 with wide margins (all three CI lower bounds also clear 0.85); only the classifier check is borderline against the stricter 0.85 bar applied to it. Source: `outputs/robustness_summary_with_ci.csv`.

weight-independent reminder that the 2–5 block is the table’s soft region (Appendix Figures 14 and 15).

5.3 Pre-registered hypotheses

Table 6 records the six hypotheses, their numeric pass criteria, and their mechanically resolved verdicts. Two are fully supported (H3, H5) and four are partially supported with named caveats (H1, H2, H4, H6). The partial verdicts surface specific methodological caveats—a near-collinear pair across two axes (C5×G1, Pearson $r = 0.965$), governance multidimensionality, one borderline robustness check (the bibliographic classifier), and a weight-sensitive bottom tier—rather than a wholesale framework failure. The C5×G1 collinearity is the largest of only five cross-axis indicator pairs at $|r| \geq 0.7$ (of 65 pairs).

5.4 Vulnerability overlay

Applying the per-vector weights (Table 2) to the literature axis scores yields per-country preparedness scores on each attack vector (Figure 6; full ranks in Appendix Table 7). The United Kingdom and South Korea lead on all three vectors; the lower tier (Israel, Taiwan, India) is least prepared, with vector-specific reordering—India is weakest on cyber and influence operations, Taiwan weakest on CBRN. Per-vector scores are signed z -scale quantities (positive = above sample-mean preparedness) and are interpretable only within a vector, not across vectors.

5.5 Counterfactual policy leverage

To operationalise the paper’s §4 claim that tractability depends on axis position, we recompute the composite under four direct policy actions per country (e.g. joining the AISI Network, publishing a safety-provisioned AI strategy, joining the Australia Group). Figure 7 reports the composite change; the highest-leverage action differs by country and by starting position. For example, with Singapore selected, joining the Australia Group raises its composite from 1.767 to 1.879 (rank 6 to 4). For

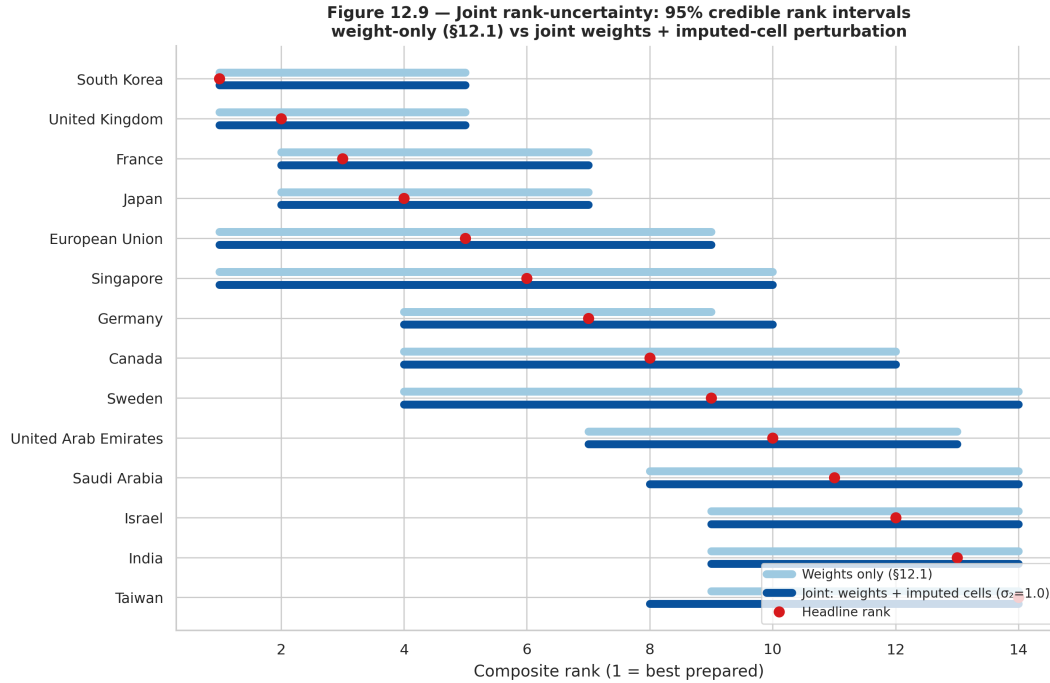


Figure 5: Joint rank-uncertainty: 95% credible rank intervals from 10,000 draws, weight-only (Section 4, §12.1) versus joint weights-plus-imputed-cell perturbation ($\sigma_2 = 1.0$); red markers are the headline ranks. Observed Tier-1 values are held fixed; only imputed cells are resampled. Source: `outputs/rank_uncertainty.csv`.

countries at an axis minimum, the geometric-shift artefact can understate the composite change, so an axis-level reading is reported alongside.

5.6 Minimal-action-set optimisation

The counterfactuals above move one lever at a time; the working paper’s §4 *tractability* claim is sharper—which improvements are *reachable*, and at what effort. Enumerating the full $2^4 = 16$ action space per country (counting only *effective* actions, so a country already in the AISI Network pays nothing to “join” it), we identify each below-top country’s *minimal action set*: the fewest effective actions that lift it into the next composite-rank tier (bottom-3 → middle → top-5; Figure 8). Tractability is starkly heterogeneous. Only two of the nine below-top jurisdictions can cross into the next tier through these four levers at all: Singapore needs a single action (join the Australia Group: rank 6 → 4) and Israel a single action (publish a safety-provisioned AI strategy: rank 12 → 10). The remaining seven—Germany, Canada, Sweden, the UAE, Saudi Arabia, India, and Taiwan—cannot reach the next tier through these governance/capacity levers alone; India and Taiwan in particular are held down by infrastructure and capacity deficits the four actions do not address. This is an explicit, per-country statement of *where* preparedness is cheaply tractable and where it is not—exactly the question §4 raises. Cost here is action count; a weighted-cost variant is a drop-in substitution.

Table 6: The six pre-registered hypotheses (criteria fixed before estimation) and their mechanically resolved verdicts.

ID	Hypothesis	Verdict	Evidence
H1	The three axes are empirically separable.	Partial	5 of 65 cross-axis pairs at $ r \geq 0.7$; C5×G1 ($r = 0.965$) is the largest contributor.
H2	Each axis carries a one-dimensional signal.	Partial	Infrastructure $p = 0.014$, governance $p = 0.040$ (both significant); capacity untestable at $n = 8$; governance PC1 explains 45.6%.
H3	The 14 powers do not cluster homogeneously.	Supported	Silhouette 0.422 at $k = 3$; ARI = 1.0 vs all three linkages (Ward, complete, average); MDS distance-rank $\rho = 0.911$.
H4	The ranking is robust to perturbation.	Partial	Equal-vs-PCA, equal-vs-literature, and normalisation clear the 0.85 bar (CI lower bounds 0.877/0.877/0.861); only the classifier (CI lower bound 0.845) sits below it.
H5	The top-5 is not driven by any one coded indicator.	Supported	Top-5 set unchanged dropping either G1 or C5; judgment-collapse min $\rho = 0.996$, max single-country shift 1.
H6	Top-5 and bottom-3 tier membership are stable.	Partial	5/5 top-5 members in tier in $\geq 70\%$ of draws; no bottom-3 member reaches 80%, Israel only 49.4%, cycling with Saudi Arabia.

5.7 Preparedness-checklist readiness

The counterfactuals and minimal-action sets above exercise four direct policy levers. The working paper’s §4, however, is a nine-action preparedness checklist—Detection (foresight, monitoring, verification), Escalation (severity schema, mandatory reporting, cross-sectoral pathways), and Mitigation & Containment (emergency channels, defensive capabilities, contingency/kill-switch)—that tags each action with the *enablers* it requires, arguing that “the tractability of these actions depends upon a country’s position within the 3 axes.” We operationalise that general claim with an *enabler-readiness matrix* (Figure 9): each checklist action’s paper-assigned enablers are transcribed verbatim, each enabler is mapped to the M-PAPI indicators (or whole axis) that proxy it through an analyst-coded crosswalk shipped as an auditable ledger (`outputs/enabler_crosswalk_ledger.csv`), and a per-country readiness score is the mean of the country’s literature-scheme z -scores on the *proxied* enablers, reported alongside a coverage fraction.

The matrix makes the paper’s heterogeneity claim concrete: top-tier countries (UK, Korea, France, EU, Japan) read broadly green because the enablers each action needs fall on axes where they already score highly, whereas capacity- or infrastructure-thin countries (India, Taiwan) are green only on the actions whose enablers sit on their stronger axis. Crucially, two of the paper’s enablers—*inter-disciplinary expertise* and *intra-governmental coordination*—have no faithful country-level indicator; they are reported as *un-proxied* (hatched in Figure 9, lowering the affected actions’ coverage) rather than imputed, so the layer neither overstates coverage nor fabricates a score it cannot support.

5.8 Causal-interpretation boundary

The counterfactuals (§5.6 and Figure 7) are *mechanical recomputations of the index* under a hypothetical indicator change—they answer “if indicator X took value x' , the composite would

Figure 15.3 — Vulnerability overlay per attack vector (paper §2)

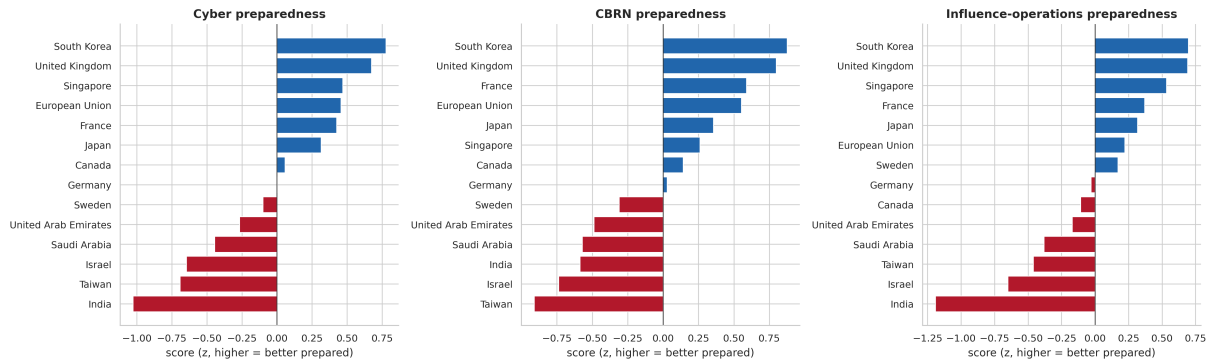


Figure 6: Per-vector preparedness (paper §2): cyber, CBRN, and influence operations. Each panel is sorted independently; bars are signed z -scale scores (higher = better prepared). Source: `outputs/vulnerability_overlay.csv`.

move by Δ —not estimates of the causal effect of a policy on real-world preparedness, and we do not present them as such. Reading a counterfactual Δ causally would require, in the language of Pearl’s do-calculus (Pearl, 2009) and the potential-outcomes framework (Hernán and Robins, 2020), a well-defined intervention, no unmeasured confounding, no interference between units, temporal precedence, and structural stability—none of which a single cross-sectional snapshot of $n = 14$ jurisdictions satisfies: AISI membership and strategy comprehensiveness co-vary with latent state capacity (the C5×G1 $r = 0.965$ signal), one country’s action shifts others’ *relative* ranks, and cause and outcome are observed simultaneously (Figure 10; the full assumptions ledger is in `outputs/causal_assumptions.csv`). The counterfactuals are therefore index-structural sensitivity probes and tractability illustrations, consistent with §4’s qualitative framing, while the paper’s §3 trigger-event/causal claims remain explicitly out of scope pending country-year incident data (Section 7). The preparedness-checklist readiness matrix (§5.7) is governed by the same boundary: it reports where each country stands on the indicators proxying each action’s required enablers, not whether the action will be executed or whether realised preparedness follows.

5.9 Typology

The silhouette-selected $k = 3$ k -means partition separates a Tier-1/asymmetric-developed group, a mid group, and India as a singleton; the partition is identical to all three hierarchical linkages (Ward, complete, and average) at ARI = 1.0. The MDS embedding (Figure 11) preserves the pairwise-distance rank at $\rho = 0.911$ and visually separates the archetypes.

6 Discussion: how the index supports the working paper

The index supplies the empirical layer the working paper asserts but does not provide.

It grounds the §1 premise. The paper claims the fourteen middle powers “differ along three axes.” H3 confirms this empirically: the countries do not cluster homogeneously. The silhouette-selected $k = 3$ k -means partition—an eight-country developed group, a five-country mixed group, and India as a singleton—is identical to all three hierarchical linkages (Ward, complete, and average) at ARI

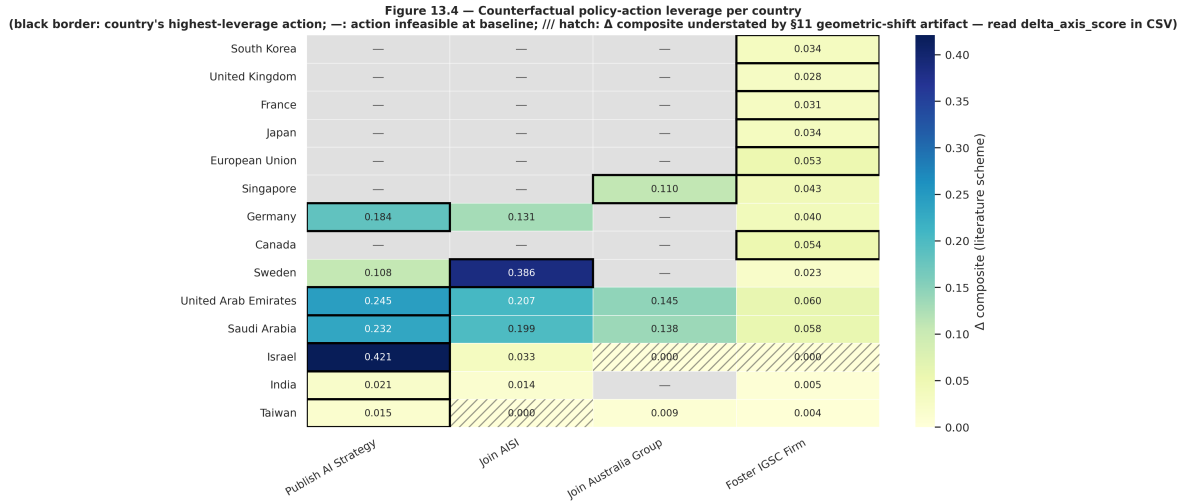


Figure 7: Counterfactual policy-action leverage per country (composite change under the literature scheme). Black border: the country’s highest-leverage action; em-dash: action already satisfied at baseline; hatching: a geometric-shift artefact for which the axis-level change should be read instead. Source: `outputs/counterfactual_scenarios.csv`.

= 1.0 (Section 5). Reading each country instead by its own axis z -profile (a per-country labelling distinct from the k -means cluster id) yields four interpretable archetypes: Tier-1 (UK, Korea, France, Japan, Singapore), capacity-leveraged (EU, Canada, Germany), infrastructure-leveraged (Sweden, UAE), and Tier-3 vulnerable (Israel, Saudi Arabia, India, Taiwan)—the archetypes colour-coded in Figures 3 and 11.

It identifies the most-exposed tier. The paper argues that lower-capacity middle powers face a “structurally under-addressed vulnerability.” The index makes that tier concrete and reproducible: India and Taiwan are the two lowest-ranked composites under all three weighting schemes, and the three least-prepared states on every per-vector overlay are India, Israel, and Taiwan (in vector-specific order); the third bottom-tier composite slot cycles between Israel and Saudi Arabia (the single-slot instability behind H6’s partial verdict). This illustrates—rather than merely asserts—which states the paper’s argument most urgently concerns.

It quantifies §4 tractability. The paper repeats that “tractability depends on a country’s position within the three axes” without specifying any action’s leverage. The counterfactual scenarios (Figure 7) convert this into a per-country, per-action quantity: the highest-leverage step differs by starting position, exactly as the paper’s qualitative argument implies. The enabler-readiness matrix (Figure 9) generalises this from the four levers to all nine checklist actions, scoring each country’s structural readiness on the indicators that proxy each action’s required enablers—and reporting honestly which enablers (inter-disciplinary expertise; intra-governmental coordination) the fourteen indicators cannot proxy.

Defensibility. Three design choices keep the study reviewable. First, *bounded scope*: the index operationalises §1–§2 and explicitly declines §3–§4. Second, *census, not sample*: the fourteen jurisdictions are the population the paper names, so the descriptive claims are exact and the small- n multivariate methods are caveated diagnostics rather than load-bearing inference. Third,

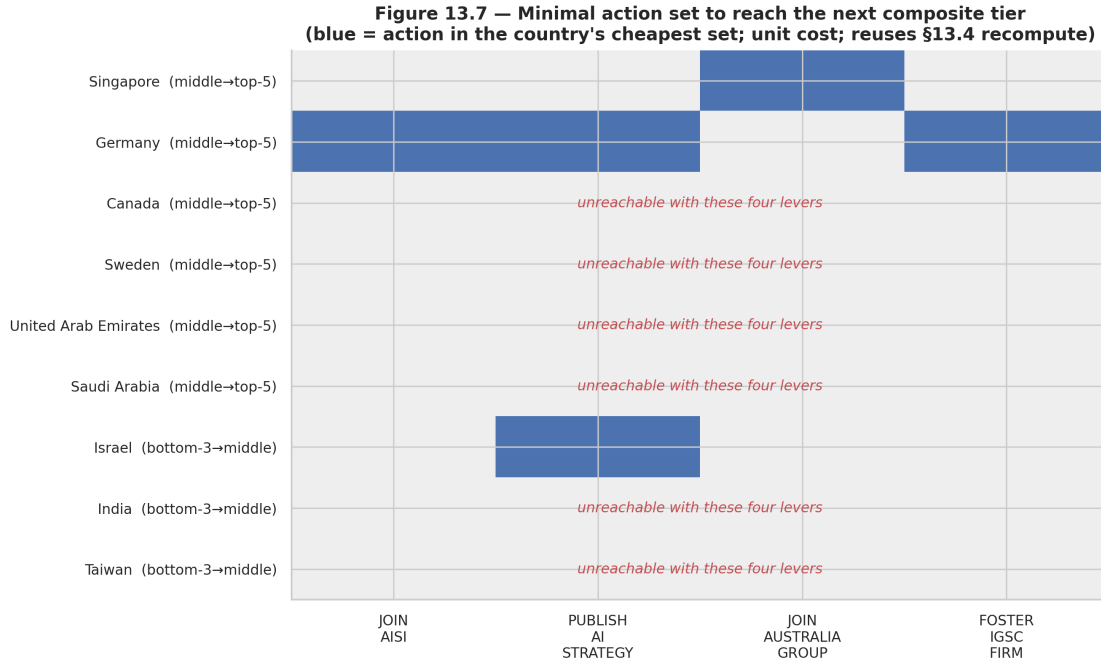


Figure 8: Minimal action set per below-top country: the fewest effective actions (blue) that reach the next composite tier; rows marked unreachable cannot cross with the four levers. Unit cost; reuses the \$13.4 recompute. Source: `outputs/minimal_action_sets.csv`.

transparency: every weakness is disclosed and bounded by a dedicated sensitivity analysis, and the partial-support verdicts are reported as such.

6.1 External concurrent validity

A composite is only as credible as its agreement with *independent* measures of the same construct. None of M-PAPI's indicators is itself a published readiness score, so the index can be benchmarked against one without circularity. The natural comparator is the Oxford Insights Government AI Readiness Index 2024 (Oxford Insights, 2024), an independently constructed index over 188 jurisdictions and 40 indicators that is *not* an input to M-PAPI. Because it scores sovereign states, it covers thirteen of the fourteen middle powers—the EU bloc is the sole exclusion, mirroring the Section 5 EU-vs-member-state treatment—and Taiwan *is* covered. Across the thirteen, the M-PAPI literature composite and the GAIRI overall score are strongly rank-concordant: Spearman $\rho = 0.852$ (95% CI [0.492, 0.963], $n = 13$; Figure 12). The agreement evidences that M-PAPI tracks a real, externally corroborated readiness gradient rather than an artefact of its own weighting. The divergences are equally informative and expected by construction: Singapore ranks first on GAIRI but sixth on the full M-PAPI ranking (fifth among the thirteen GAIRI-covered jurisdictions, with the EU excluded), and the United Kingdom and Japan rank higher on M-PAPI than on GAIRI—M-PAPI deliberately rewards the CBRN/biosecurity-governance signals (Australia Group membership, IGSC participation, AISI-network presence) that a general-purpose government-readiness index omits, which is precisely the proliferation-specific lens the working paper calls for.

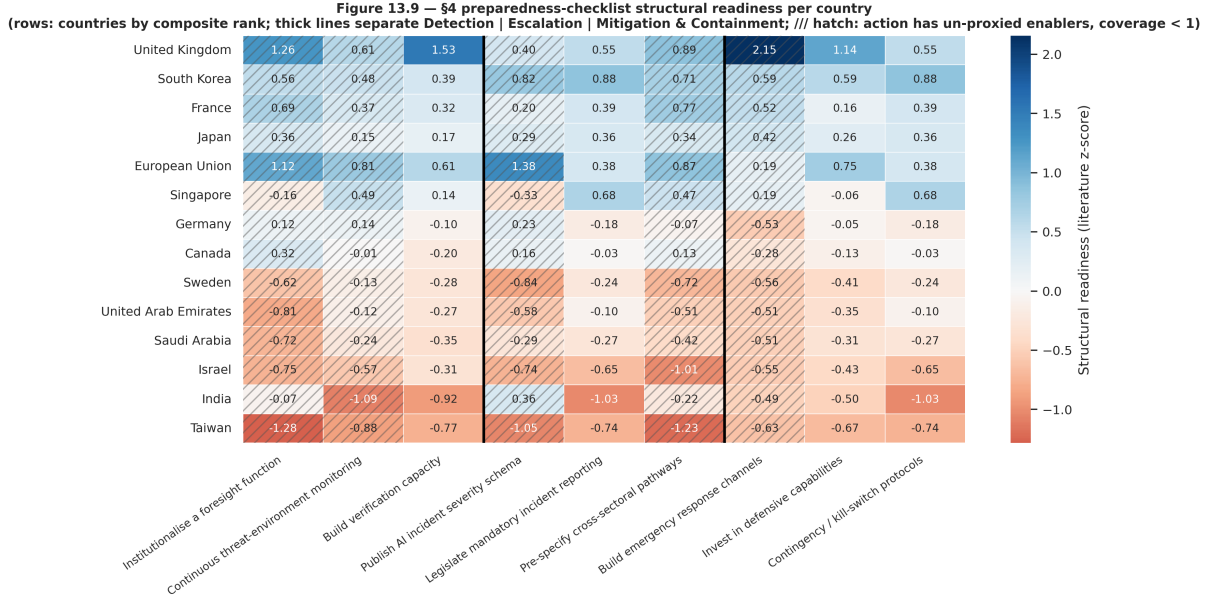


Figure 9: §4 preparedness-checklist structural readiness per country: the mean literature-scheme z -score over the indicators proxying each checklist action’s required enablers. Rows ordered by composite rank; thick rules separate Detection, Escalation, and Mitigation & Containment; hatching marks actions with un-proxied enablers (coverage < 1). A structural illustration, not a causal claim (§5.8). Source: `outputs/preparedness_readiness_matrix.csv`.

7 Limitations and threats to validity

Following Shadish et al. (2002), we organise residual risks by validity type.

Construct validity. Several indicators are disclosed proxies whose link to the axis label is indirect: ND-GAIN readiness (I5) is a climate-adaptation measure used as an infrastructure-resilience proxy; V-Dem (G3) measures political-system democracy, not AI governance *per se*; the IGSC count (G5) measures industrial biosecurity participation; and secure-server density (I3) increasingly reflects certificate-registration jurisdiction rather than substantive infrastructure. These are the most pressable choices; the defence is that the axes are the paper’s, the proxies are transparent, and the ranking is robust to dropping any single indicator. The working paper’s third axis is *Compute and Infrastructure Posture*; the headline axis is built from ICT-infrastructure measures (I1–I3, I5), and a dedicated compute indicator is sourced and tested in Appendix B (the TOP500 supercomputer-system count, attributed by installation country, so distinct from C2’s organisation-attributed training compute—the overlap that ruled out an earlier Epoch-based candidate). Adding it leaves the ranking essentially unchanged (Spearman $\rho = 0.996$; only an EU–Japan swap at ranks 4–5) but tips the silhouette-selected k to 2, where average-linkage clustering degenerates and the pre-registered H3 criterion would fail on that artefact; it is therefore reported as a robustness check rather than folded into the headline. A hosting-side performance (Rmax) variant and coverage of unsubmitted private clusters remain future work (TOP500 is voluntary-submission).

Cross-axis collinearity. C5 (AISII presence) and G1 (AI-strategy comprehensiveness) are near-collinear ($r = 0.965$) yet sit in different axes, so a shared institutional signal enters the composite twice. Reassigning C5 to governance leaves the top-five set unchanged (Spearman $\rho = 0.982$, maximum single-country shift 1), so the double-count does not re-order the table. The

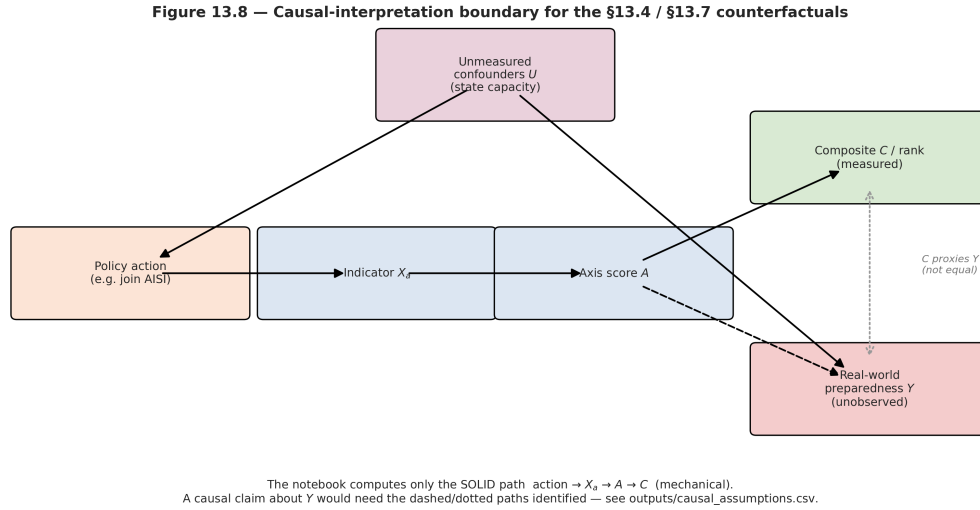


Figure 10: Causal-interpretation boundary for the counterfactuals. The index computes only the solid path (action $\rightarrow X_a \rightarrow A \rightarrow$ composite C); a causal claim about real-world preparedness Y would need the dashed/dotted paths identified, which an $n = 14$ cross-section cannot do. Source: outputs/causal_assumptions.csv.

exact second-order Shapley interaction index puts a magnitude on the overlap *inside* the composite: $I(C5, G1) = +0.009$ —small and mildly *complementary*, not substitutive. The high correlation is thus a shared-information property (the two indicators co-vary across countries) rather than a large additive redundancy in the aggregated score; because $C5$ and $G1$ sit in different axes, the cross-axis geometric mean makes their joint marginal contribution slightly positive. The genuinely substitutive interactions are all within-axis pairs (e.g. $C1 \times C2 = -0.021$), exactly as linear within-axis aggregation of correlated indicators predicts (Appendix C). The double-count remains an acknowledged construct flaw—two indicators of one latent institutional-capacity signal—but its measured aggregate effect is bounded.

Weighting and aggregation. The vector-overlay weights are authorial translations of the paper’s qualitative arguments, so the per-vector rankings are the most judgment-dependent output. The across-axis geometric mean introduces a shift constant ε : an ε -grid spanning 10^{-4} to 1 shows the ranking is ε -invariant (Spearman $\rho = 1.000$ for $\varepsilon \leq 0.01$, 0.987 at $\varepsilon = 0.1$, 0.982 at $\varepsilon = 1$), but the bottom-tier composite *values* are an ε -artefact (Taiwan’s value spans 0.024 to 1.56 across the grid). The bottom tier should therefore be cited as ranks, not magnitudes. More generally, because the geometric shift references each axis’s sample minimum, every composite *value* is defined relative to the fourteen-country frame and the worst performer within it; the *ranking* is the portable quantity, and composite magnitudes should not be read as cardinal, frame-independent preparedness levels.

Inference and external validity. With $n = 14$, point-estimate rank correlations carry non-negligible sampling variance, which the Bonett–Wright CIs make explicit. The fourteen units are also not mutually independent: the European Union row is aggregated from member states that include three of the other thirteen units (France, Germany, and Sweden), so the z -score normalisation, the PCA loadings, the clustering, and the cross-axis correlations all rest on an overlapping-units assumption that a strict inferential reading would violate. We treat the fourteen as the paper’s named census rather than a random sample (Section 6), and the leave-one-country-out jackknife bounds how much the (partly synthetic) EU row—or any single unit—drives the ranking and the typology; the

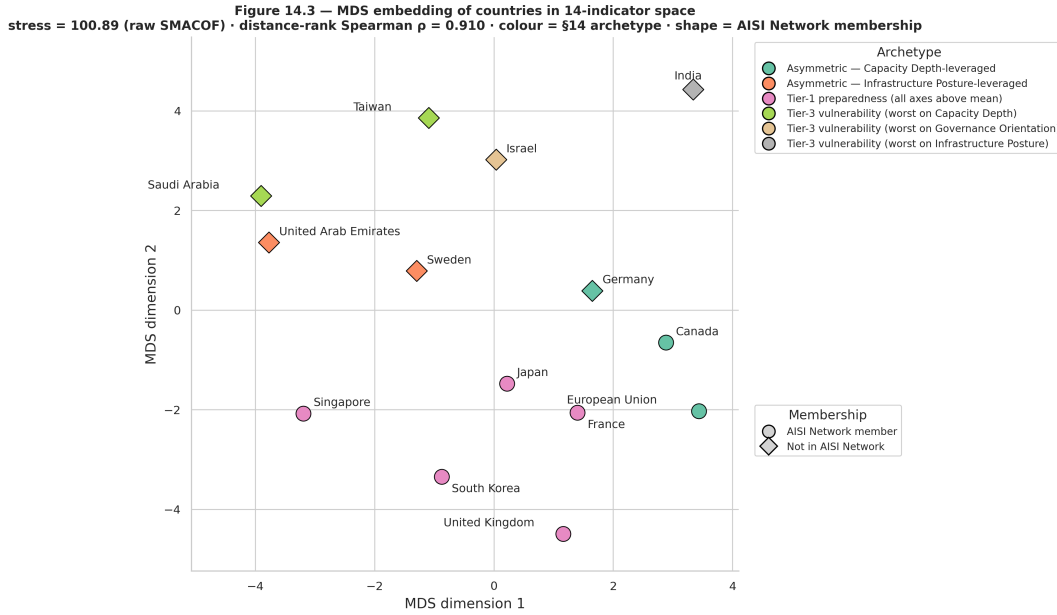


Figure 11: Multidimensional-scaling embedding of the 14 jurisdictions in the full fourteen-indicator space (distance-rank $\rho = 0.911$). Colour: archetype; marker shape: AISI Network membership. Source: `outputs/mds_coordinates.csv`.

non-independence nonetheless remains a caveat on every distributional statistic reported here. The sample is the fourteen named middle powers only; results do not generalise to non-middle-powers or small states, and composite values are not portable outside the fourteen-country reference frame. The ranking does, however, agree strongly with the independent Oxford Insights GAIIRI 2024 index over the thirteen jurisdictions both cover (Spearman $\rho = 0.852$; Section 6.1), which bounds the concern that the ordering is an artefact of the authors’ own weighting.

8 Conclusion

M-PAPI converts the working paper’s qualitative three-axes framework into a transparent, reproducible composite indicator over the paper’s own fourteen named middle powers. It empirically positions each country, confirms that the countries are differentiable, identifies the most-exposed tier, quantifies the per-country leverage of preparedness actions, and scores structural readiness across the §4 checklist—while honestly declining the trigger-event claims it cannot yet test. The headline ranking is robust to weighting and normalisation and borderline only to a single indicator-construction choice; the hypothesis verdicts (two fully, four partially supported) are reported with their specific caveats; and, within a pinned environment, every output regenerates byte-for-byte. The study is offered as auditable empirical scaffolding for the working paper, and as a template that can be re-run as the underlying data and the threat landscape evolve.

Data and code availability

The notebook, its outputs and figures, the bundled offline data snapshot, and the source of this paper are openly available at <https://github.com/khailapvuong/middle-powers-mpapi> — code

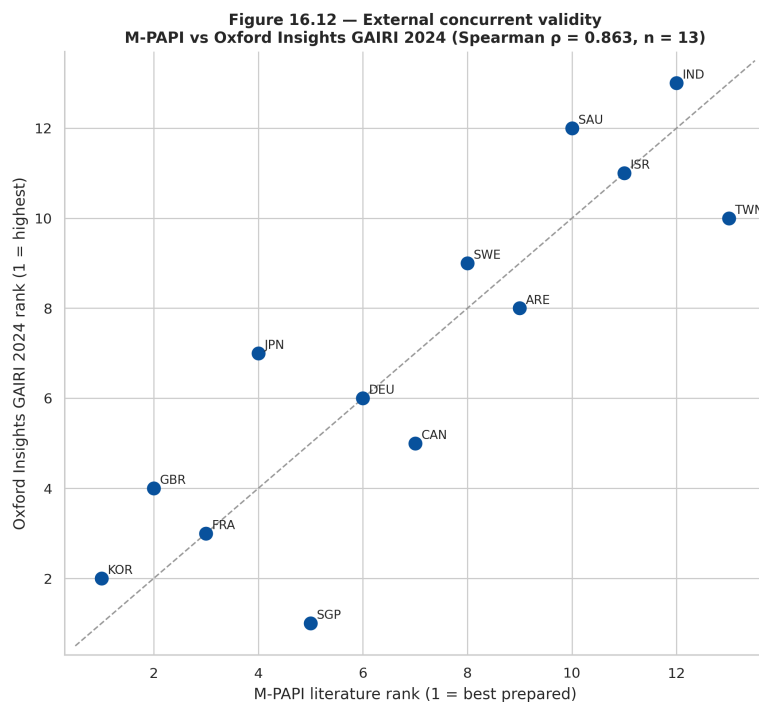


Figure 12: External concurrent validity: M-PAPI literature rank versus Oxford Insights GAIRI 2024 rank across the thirteen covered jurisdictions (the EU bloc is unscored by GAIRI). Points near the diagonal agree; Spearman $\rho = 0.852$. Source: `outputs/external_validity.csv`.

under the MIT licence, bundled data under the respective upstream licences listed in Appendix A. All results regenerate from the bundled snapshot offline (MPAPI_OFFLINE=1; Python 3.11+, `pip install -r requirements.txt`); a pinned `requirements.lock` reproduces the figures byte-for-byte, and a continuous-integration workflow re-executes the notebook on every commit and fails on any discrepancy with the documented hypothesis verdicts. Two source reports under restrictive redistribution terms (the ITU ICT Development Index report and the Stanford AI Index chapter) are not redistributed; the derived indicator values, the extraction scripts, and the source URLs are provided in their place.

Author contributions

All authors developed the three-axes framework operationalised here in the parent working paper. The composite-indicator implementation, the sensitivity and robustness analysis, the reproducibility pipeline, and this companion paper were prepared by the author team; all authors reviewed and approved the final manuscript.

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Competing interests

The authors declare no competing interests.

References

- Australia Group. Australia Group Participants, 2024. URL <https://www.australiagroup.net/en/participants.html>.
- Douglas G. Bonett and Thomas A. Wright. Sample size requirements for estimating Pearson, Kendall and Spearman correlations. *Psychometrika*, 65(1):23–28, 2000. doi: 10.1007/BF02294183.
- Michael Coppedge, John Gerring, Carl Henrik Knutsen, et al. V-Dem Country-Year Dataset v16. Varieties of Democracy (V-Dem) Project, 2026. URL <https://www.v-dem.net/>.
- Lee J. Cronbach. Coefficient alpha and the internal structure of tests. *Psychometrika*, 16(3):297–334, 1951. doi: 10.1007/BF02310555.
- Jean-Charles de Borda. Mémoire sur les élections au scrutin. In *Histoire de l’Académie Royale des Sciences*. Imprimerie Royale, Paris, 1781.
- Alexandra de Hoop Scheffer, Martin Quencez, Rachel Tausendfreund, Garima Mohan, Gabriel Weber, Sudha Jagtiani, and Kristina Kausch. Pivotal Powers. Technical report, German Marshall Fund of the United States, 2024. URL <https://www.gmfus.org/>.
- Epoch AI. Data on Notable AI Models. <https://epoch.ai/data>, 2024. Licence CC-BY 4.0. Retrieved 2026-05-06.
- Michel Grabisch and Marc Roubens. An axiomatic approach to the concept of interaction among players in cooperative games. *International Journal of Game Theory*, 28(4):547–565, 1999. doi: 10.1007/s001820050125.
- Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, et al. Array programming with NumPy. *Nature*, 585:357–362, 2020. doi: 10.1038/s41586-020-2649-2.
- Miguel A. Hernán and James M. Robins. *Causal Inference: What If*. Chapman & Hall/CRC, Boca Raton, 2020.
- John L. Horn. A rationale and test for the number of factors in factor analysis. *Psychometrika*, 30(2):179–185, 1965. doi: 10.1007/BF02289447.
- Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, 1985. doi: 10.1007/BF01908075.
- John D. Hunter. Matplotlib: A 2d graphics environment. *Computing in Science & Engineering*, 9(3):90–95, 2007. doi: 10.1109/MCSE.2007.55.
- International Gene Synthesis Consortium. IGSC Member Roster, 2026. URL <https://genesynthesiconsortium.org/>.
- International Telecommunication Union. Global Cybersecurity Index 2024. Technical report, ITU, 2024a. 5th edition; accessed via World Bank Data360.

- International Telecommunication Union. Measuring digital development: The ICT Development Index 2024. Technical report, ITU, 2024b. URL <https://www.itu.int/en/ITU-D/Statistics/>.
- John G. Kemeny. Mathematics without numbers. *Daedalus*, 88(4):577–591, 1959.
- Joseph B. Kruskal. Multidimensional scaling by optimizing goodness of fit to a nonmetric hypothesis. *Psychometrika*, 29(1):1–27, 1964. doi: 10.1007/BF02289565.
- Nestor Maslej, Loredana Fattorini, Raymond Perrault, et al. The AI Index 2025 Annual Report. Technical report, Stanford Institute for Human-Centered Artificial Intelligence (HAI), 2025. URL <https://hai.stanford.edu/ai-index/2025-ai-index-report>. Chapter 1, Figure 1.2.3 (AI patents per 100,000 inhabitants).
- Wes McKinney. Data structures for statistical computing in Python. In *Proceedings of the 9th Python in Science Conference*, pages 56–61, 2010. doi: 10.25080/Majora-92bf1922-00a.
- OECD. OECD.AI Policy Observatory: National AI Policies Database, 2024. URL <https://oecd.ai/en/dashboards/national>.
- OECD and European Commission, Joint Research Centre. *Handbook on Constructing Composite Indicators: Methodology and User Guide*. OECD Publishing, Paris, 2008. doi: 10.1787/9789264043466-en.
- Oxford Insights. Government AI Readiness Index 2024. Technical report, Oxford Insights, 12 2024. URL <https://oxfordinsights.com/wp-content/uploads/2024/12/2024-Government-AI-Readiness-Index-2.pdf>. 188 jurisdictions; 40 indicators across the Government, Technology Sector, and Data & Infrastructure pillars. Retrieved 2026-06-12.
- Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, Cambridge, 2 edition, 2009.
- Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- Jason Priem, Heather Piwowar, and Richard Orr. OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. *arXiv preprint arXiv:2205.01833*, 2022. doi: 10.48550/arXiv.2205.01833.
- Maurice H. Quenouille. Approximate tests of correlation in time-series. *Journal of the Royal Statistical Society, Series B*, 11(1):68–84, 1949.
- Peter J. Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987. doi: 10.1016/0377-0427(87)90125-7.
- Felipe J. V. Sandoval, Isabella Wilkinson, Alex Krasodonski, and Rowan Wilkinson. How middle powers can weather US and Chinese AI dominance: The case for ‘sovereign AI’ strategies. Technical report, Royal Institute of International Affairs (Chatham House), 2026. URL <https://doi.org/10.55317/9781784136710>.
- William R. Shadish, Thomas D. Cook, and Donald T. Campbell. *Experimental and Quasi-Experimental Designs for Generalized Causal Inference*. Houghton Mifflin, Boston, 2002.

- Lloyd S. Shapley. A value for n -person games. In Harold W. Kuhn and Albert W. Tucker, editors, *Contributions to the Theory of Games II*, pages 307–317. Princeton University Press, 1953.
- James Teague, Alina Ali, Sophie Sfeir, and Kristina Fort. AI-Proliferation and Middle Powers: Preparation and Response Mechanisms. Working paper. Companion notebook: <https://github.com/khailapvuong/middle-powers-mpapi>, 2026.
- TOP500.org. TOP500 list — november 2025 (66th edition). <https://www.top500.org/lists/top500/2025/11/>, 2025. Accessed 2026-06-13.
- Tortoise Media. The Global AI Index. Tortoise Intelligence, 2023. URL <https://www.tortoisemedia.com/intelligence/global-ai/>.
- John W. Tukey. Bias and confidence in not-quite large samples. *Annals of Mathematical Statistics*, 29(3):614, 1958.
- University of Notre Dame. Notre Dame Global Adaptation Initiative (ND-GAIN) Country Index, 2026. URL <https://gain.nd.edu/our-work/country-index/>.
- U.S. Department of Commerce. Fact Sheet: U.S. Department of Commerce & U.S. Department of State Launch the International Network of AI Safety Institutes. NIST, 2024. URL <https://www.nist.gov/news-events/news/2024/11/fact-sheet-us-department-commerce-us-department-state-launch-international>.
- Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.
- Jenny Wang, Chloe Milne, Zhang Jess Hu, and Farah Khan. Navigating geopolitics in AI governance. Technical report, Oxford Global Society, 2024. URL <https://oxgs.org/2024/04/08/oxgs-report-navigating-geopolitics-in-ai-governance/>.
- Michael L. Waskom. seaborn: statistical data visualization. *Journal of Open Source Software*, 6(60):3021, 2021. doi: 10.21105/joss.03021.
- World Bank. World Development Indicators, 2024. URL <https://data.worldbank.org/>. Series IT.NET.BBND.P2 (fixed broadband) and IT.NET.SECR.P6 (secure servers). Licence CC-BY 4.0.
- H. Peyton Young. Condorcet’s theory of voting. *American Political Science Review*, 82(4):1231–1244, 1988. doi: 10.2307/1961757.

A Data dictionary and provenance

Every indicator value cites a named source with a retrieval date and URL, recorded in per-file `*.meta.json` provenance sidecars. Programmatic API sources were retrieved on 2026-05-06; the PDF/HTML-derived sources (Stanford patents, ITU IDI, IGSC roster) on 2026-05-18. The bundled `data/raw/` snapshot makes the analysis reproducible offline. Source licences: Epoch AI (CC-BY 4.0), World Bank (CC-BY 4.0), OpenAlex (CC-0), V-Dem (CC-BY), ND-GAIN (free with attribution), ITU (citable academic use), Stanford HAI (free with attribution), NIST AISI Network (U.S. public domain), IGSC and Australia Group (public rosters).

B Full robustness and dimensionality

Table 7 gives the full per-vector vulnerability ranking. Figure 16 gives the Monte Carlo rank distributions; Figure 17 the within-axis indicator correlations; Figure 18 the full indicator correlation matrix; Figure 19 the hierarchical clustering under three linkage methods; Figures 20, 21, and 22 the axis scoreboard, pairwise axis scatter matrix, and sparse-PCA governance sub-axes respectively; Figures 14 and 15 the leave-one-country-out jackknife and the weighting-free consensus ranking; and Figure 13 the TOP500 compute-indicator robustness check.

The ε -sensitivity, C5-reassignment, and analyst-coded judgment-collapse checks all leave the top-five set intact: the analyst-collapse variants (G1 and C5 individually and both at once) return min Spearman $\rho = 0.996$ with at most a single mid-table shift, and the C5-reassignment returns $\rho = 0.982$. Swapping the across-axis geometric mean for an arithmetic mean of the same min-shifted axis scores returns $\rho = 0.978$ (95% CI [0.911, 0.995], maximum single-country shift two), with the top three invariant and the movement concentrated in the middle and lower tier where the geometric min-shift floor bites. Cross-axis correlations at $|r| \geq 0.7$ number five of sixty-five pairs (C5×G1 0.965; C1×G1 0.781; C4×I3 −0.748; G3×I3 0.722; C3×G6 0.702). Tier-membership stability over 10,000 Monte Carlo draws: all five top-5 members in tier $\geq 70\%$ of draws (UK 98.6%, Korea 97.9%, France 86.8%, Japan 83.3%, EU 76.8%); no bottom-3 member reaches 80% (India 75.0%, Taiwan 60.9%, Israel 49.4%); exact-set-match 51%/23% and within-one-swap 93%/66% for the top-5/bottom-3.

The leave-one-country-out jackknife (Figure 14) and the weighting-free consensus ranking (Figure 15) corroborate the headline: dropping any single jurisdiction moves the retained thirteen by at most three rank positions (retained-order Spearman $\rho \geq 0.945$; typology ARI = 1.0 for twelve of fourteen drops, falling to 0.61 (India) and 0.47 (Taiwan) for the other two), and a consensus that uses no cardinal weights tracks the literature ordering at Borda $\rho = 0.977$ and Kemeny–Young $\rho = 0.965$.

Compute-indicator robustness. The working paper’s third axis is named *Compute and Infrastructure Posture*; we source a dedicated, hosting-attributed compute indicator—each country’s count of TOP500 supercomputers (TOP500.org, 2025) (66th list, November 2025, aggregated by installation country, so distinct from C2’s organisation-attributed training compute)—and test folding it into the infrastructure axis (Figure 13). It is well-populated across all fourteen (minimum: Israel, three systems; no zeros). Adding it leaves the headline essentially unchanged: Spearman $\rho = 0.996$ against the literature ranking (95% CI [0.981, 0.999]), maximum single-country shift one, with only an EU↔Japan swap at ranks 4–5. We report it as a robustness check rather than folding it into the headline because doing so tips the silhouette-selected k from 3 to 2, at which the k -means partition matches Ward and complete linkage exactly (ARI = 1.0) but average linkage degenerates to a single-outlier (India) split (ARI ≈ 0.05), mechanically failing the pre-registered H3 min-ARI criterion despite strong clustering. The compute dimension is thus sourced, validated, and disclosed while the pre-registered headline stays stable.

C Interpretability

Figure 23 decomposes each country’s composite into exact Shapley contributions per indicator (all 2^{14} coalitions; additivity verified to 10^{-6}). Figure 24 reports permutation feature importance for the literature ranking. Training compute (C2) and ITU GCI (G2) are the most rank-disruptive indicators; secure servers (I3) and ND-GAIN readiness (I5) the least. Figure 25 reports the exact

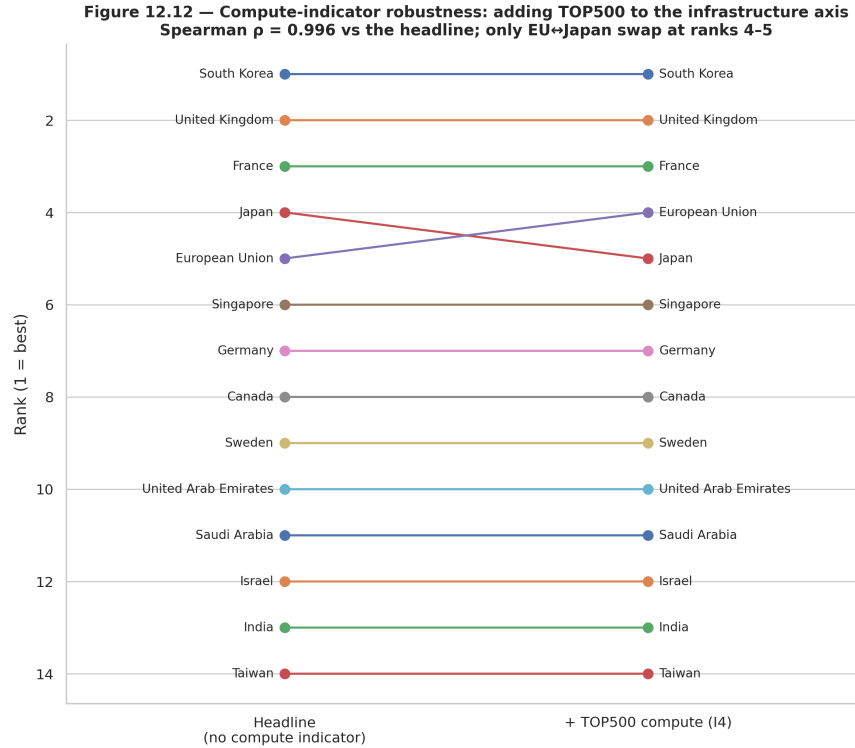


Figure 13: Compute-indicator robustness: the headline ranking versus the ranking with a TOP500 compute indicator (I4) added to the infrastructure axis. The only change is an EU↔Japan swap at ranks 4–5 (Spearman $\rho = 0.996$ vs the headline). Source: `outputs/compute_indicator_sensitivity.csv`.

second-order Shapley interaction indices (Grabisch and Roubens, 1999) for every indicator pair, averaged over the fourteen countries: the within-axis pairs are the substitutive (negative) ones, while cross-axis pairs—including the much-discussed C5×G1 (+0.009)—are weakly complementary, locating indicator redundancy firmly within rather than across axes.

D Reproducibility

All eight stochastic steps are seeded (`SEED = 20260506`): the Monte Carlo weights, Horn’s parallel-analysis null, the joint rank-uncertainty Monte Carlo, the per-vector vulnerability-overlay Monte Carlo, *k*-means, MDS, sparse PCA, and permutation importance. The leave-one-country-out re-clustering (§5.6 companion in §4) reuses the same seeded *k*-means; the Borda count and minimal-action-set enumeration are exact, and the Kemeny–Young median ranking is solved exactly by `scipy.optimize.milp` (HiGHS), deterministic within the pinned environment. With a pinned environment and the offline data snapshot (`MPAPI_OFFLINE=1`), the forty output files (thirty-seven CSV, three JSON) and all twenty-five figures reproduce byte-for-byte across successive executions, verified by hashing two consecutive offline runs. The notebook additionally reconstructs one country’s composite end-to-end (asserted within 10^{-3}) and verifies Shapley additivity (within 10^{-6}). The analysis is built on the open scientific Python stack (Harris et al., 2020; McKinney, 2010; Virtanen et al., 2020; Pedregosa et al., 2011; Hunter, 2007; Waskom, 2021).

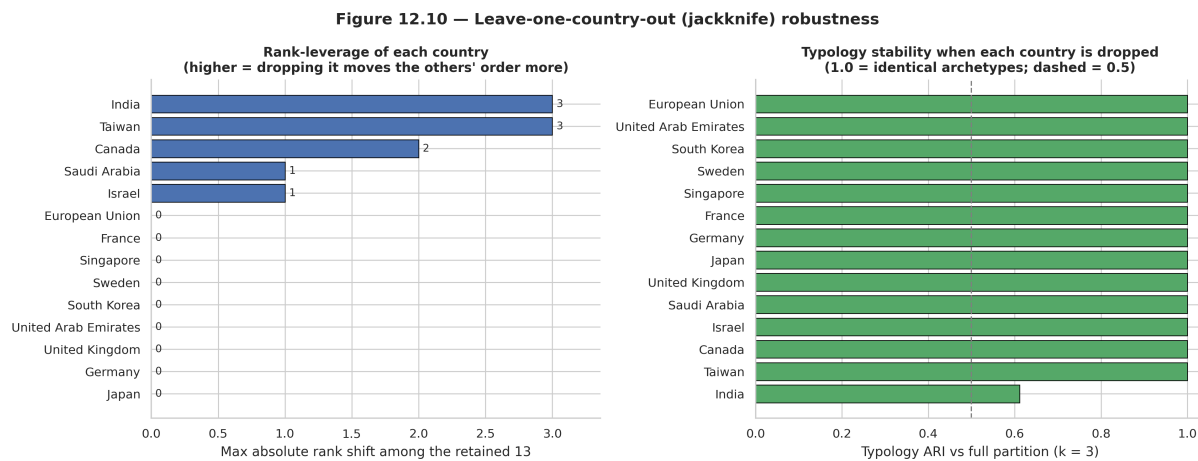


Figure 14: Leave-one-country-out (jackknife) robustness. Left: the maximum absolute rank shift among the retained thirteen when each country is dropped (rank-leverage). Right: the Adjusted Rand Index between the refit $k = 3$ typology and the full partition. Source: `outputs/leave_one_country_out.csv`.

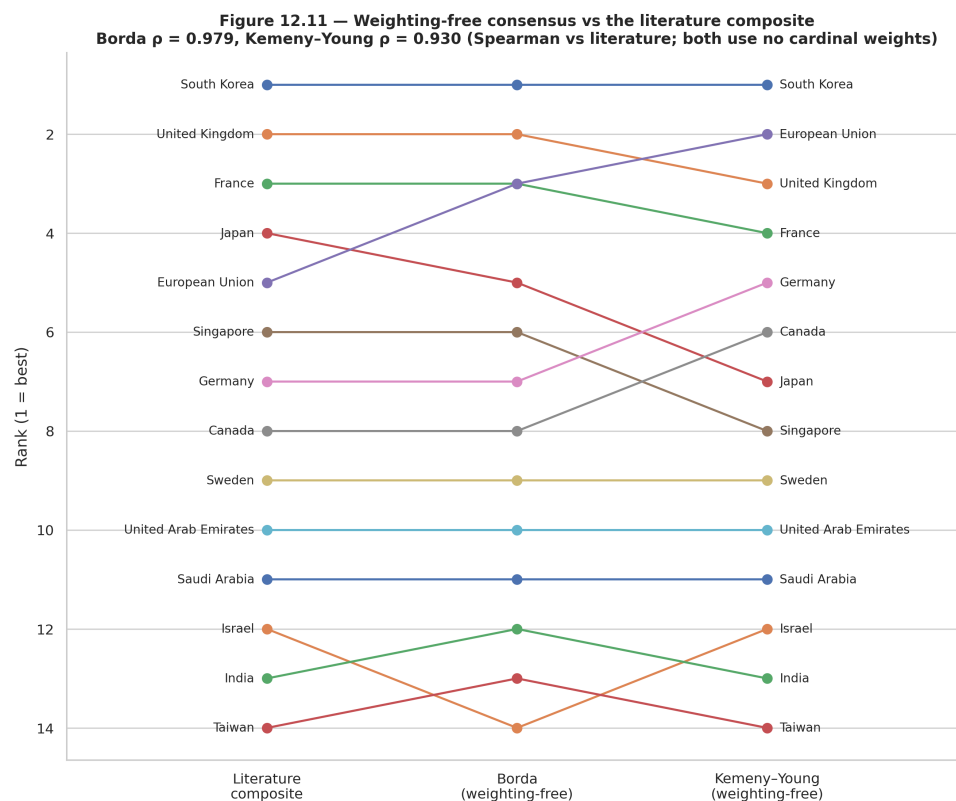


Figure 15: Weighting-free consensus ranking. The literature composite rank versus the Borda count and the exact Kemeny–Young median ranking, each aggregating the fourteen indicator ballots with no cardinal weights. Source: `outputs/rank_aggregation.csv`.

Table 7: Per-vector preparedness scores and ranks. Higher score = better prepared; signed z -scale, interpretable within a vector only. Source: `outputs/vulnerability_overlay.csv`.

Country	Cyber	Rk	CBRN	Rk	Infl.	Rk
United Kingdom	0.71	1	0.84	1	0.71	1
South Korea	0.70	2	0.77	2	0.65	2
Singapore	0.50	3	0.30	6	0.55	3
European Union	0.48	4	0.58	4	0.23	6
France	0.46	5	0.62	3	0.38	4
Japan	0.32	6	0.36	5	0.32	5
Canada	0.01	7	0.08	7	-0.13	9
Germany	-0.02	8	0.01	8	-0.04	8
Sweden	-0.11	9	-0.32	9	0.17	7
United Arab Emirates	-0.27	10	-0.49	10	-0.17	10
Saudi Arabia	-0.43	11	-0.55	11	-0.37	11
Israel	-0.63	12	-0.72	13	-0.64	13
Taiwan	-0.69	13	-0.91	14	-0.46	12
India	-1.02	14	-0.58	12	-1.19	14

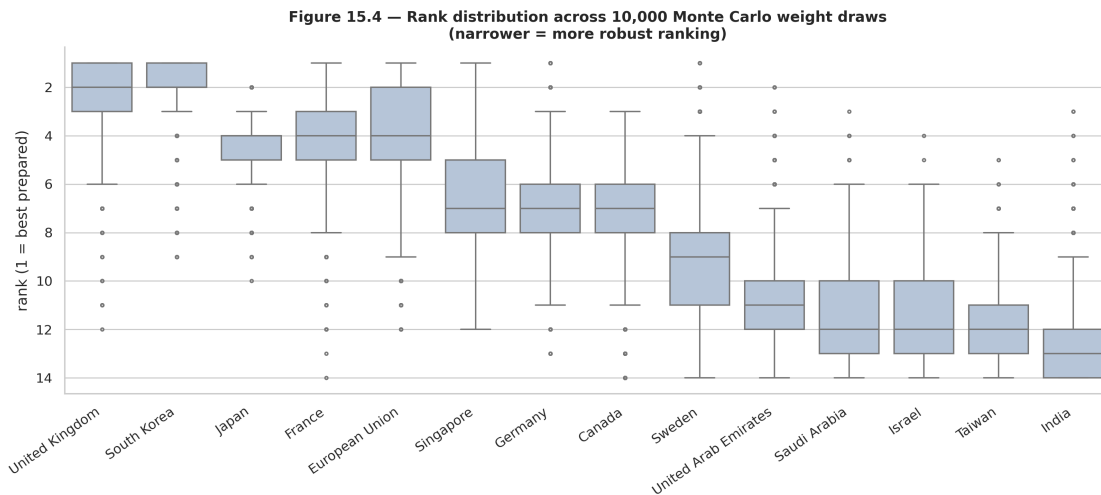


Figure 16: Per-country rank distribution across 10,000 Monte Carlo weight draws (narrower = more robust). Source: `outputs/sensitivity_ranks.csv`.

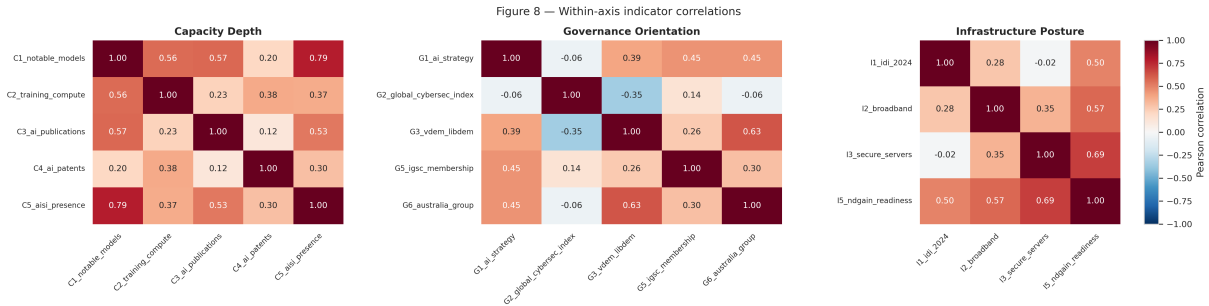


Figure 17: Within-axis indicator correlations (Pearson) for the three axes, with a shared colour scale. The governance panel shows the weak/negative association of the ITU GCI (G2) with its axis-mates that underlies the governance axis's lower internal consistency. Source: `outputs/cross_axis_correlations.csv`.

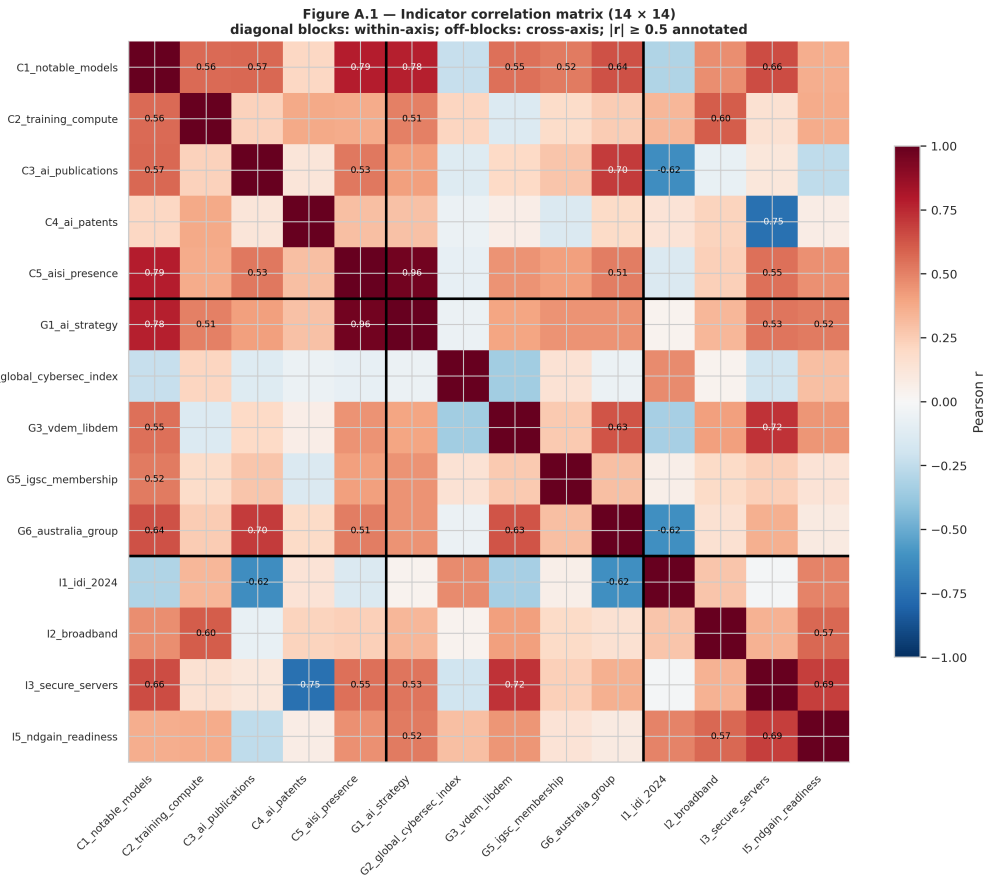


Figure 18: Full 14 × 14 indicator correlation matrix with axis-block separators; $|r| \geq 0.5$ annotated. Diagonal blocks are within-axis; off-blocks are cross-axis. Source: `outputs/cross_axis_correlations.csv`.

Figure 14.2 — Hierarchical clustering at $k = 3$ under three linkage methods

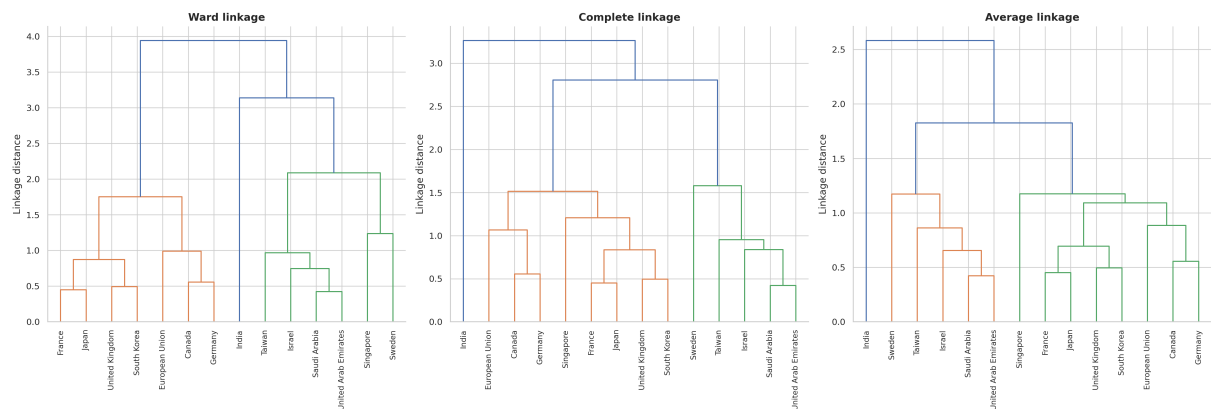
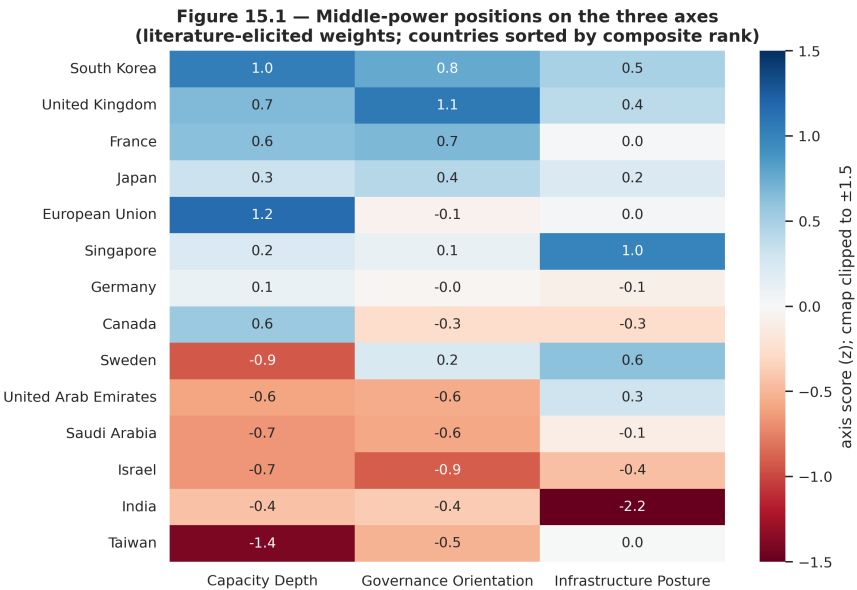


Figure 19: Hierarchical clustering at $k = 3$ under Ward, complete, and average linkage. All three linkage partitions (Ward, complete, average) are identical to the k -means partition ($\text{ARI} = 1.0$). Source: `outputs/hierarchical_ari.csv`.



Taiwan's Infrastructure Posture cell is column-mean-imputed (its four infrastructure indicators are unobserved; see §14 observability caveat).

Figure 20: Per-country axis scores under the literature scheme (z -scale, clipped to ± 1.5), ordered by composite rank. Taiwan's infrastructure cell is column-mean-imputed. Source: `outputs/index_baseline.csv`.

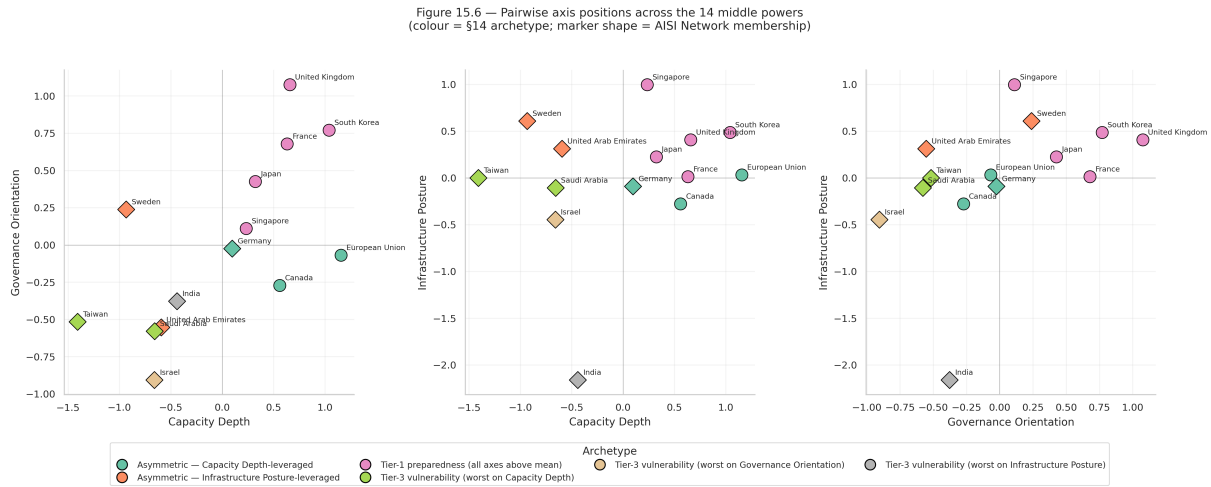


Figure 21: Pairwise axis positions across the 14 middle powers; colour: archetype; marker shape: AISI Network membership. Source: `outputs/index_baseline.csv`.

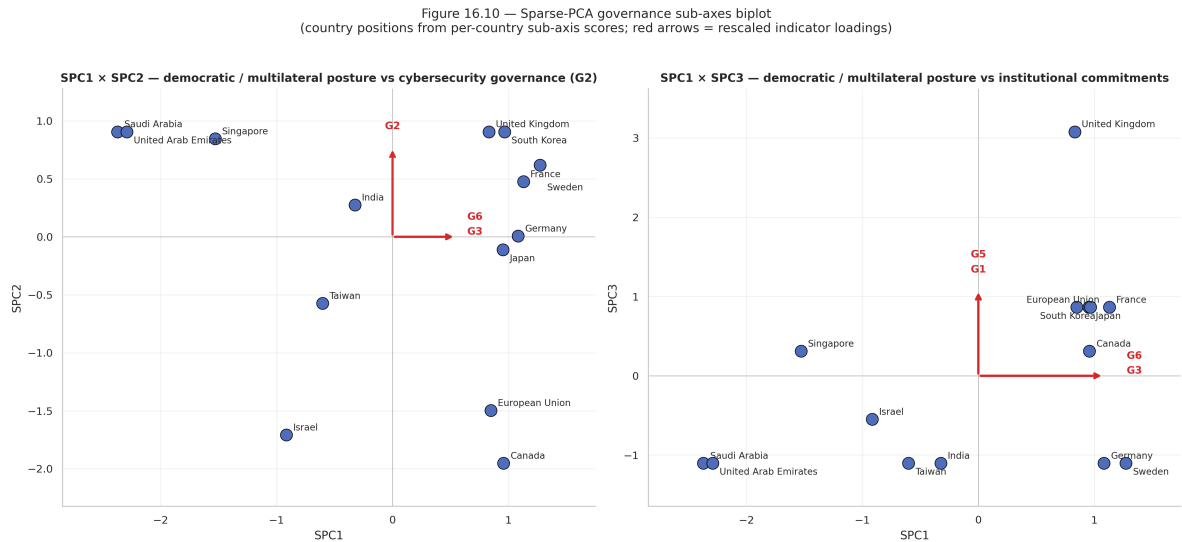


Figure 22: Sparse-PCA governance sub-axes biplot: country positions with rescaled indicator loadings (red arrows), decomposing the governance axis into institutional-commitment, democratic/multilateral, and cybersecurity sub-axes. Source: `outputs/governance_subaxes.csv`.

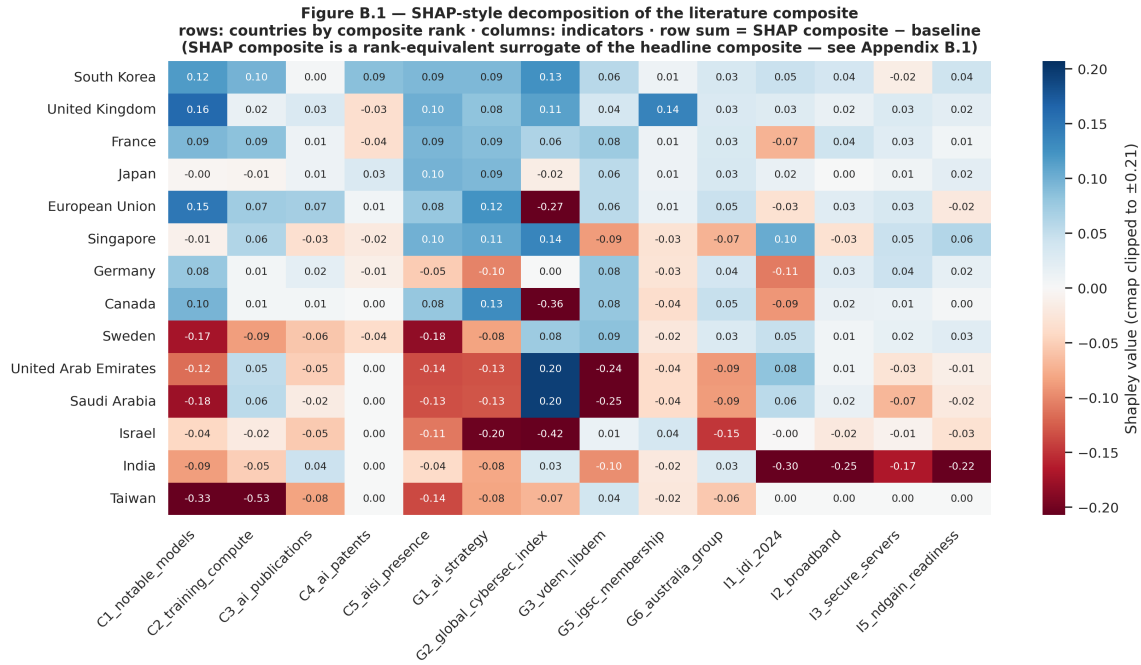


Figure 23: Exact Shapley decomposition of the literature composite. Rows: countries by composite rank; columns: indicators; each row sums to the composite minus the universal baseline. Source: `outputs/shap_attributions.csv`.

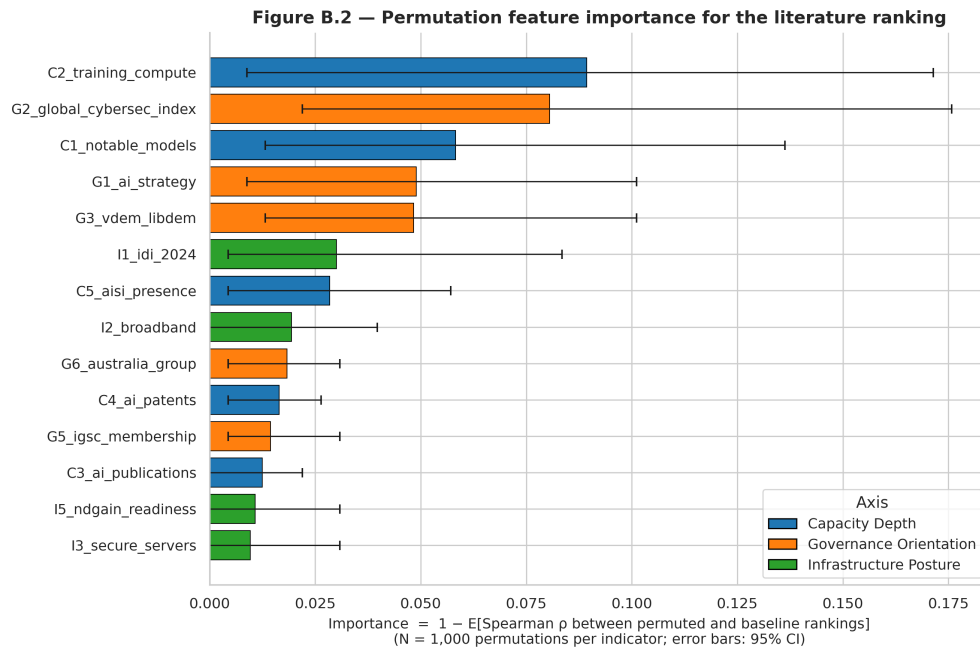


Figure 24: Permutation feature importance for the literature ranking (importance = $1 - \mathbb{E}[\text{Spearman } \rho]$ between permuted and baseline rankings; 1,000 permutations per indicator), coloured by axis. Source: `outputs/permutation_importance.csv`.

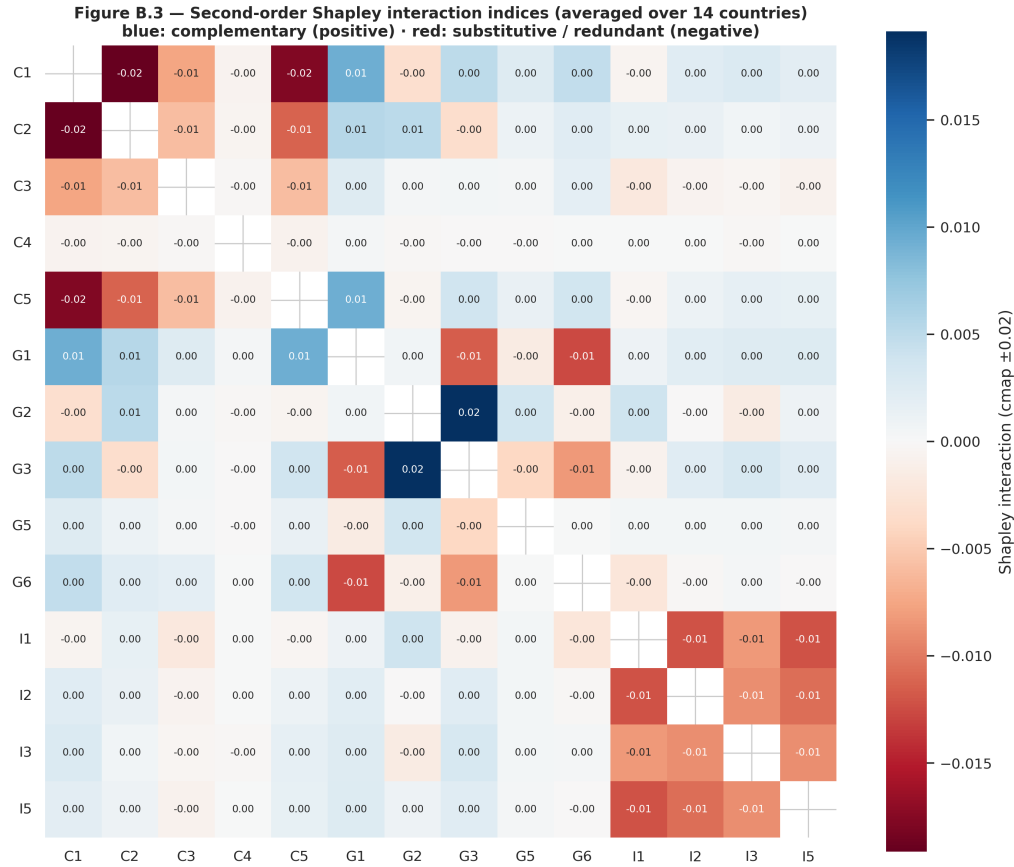


Figure 25: Second-order Shapley interaction indices for every indicator pair, averaged over the fourteen countries (all 2^{14} coalitions; [Grabisch and Roubens, 1999](#)). Blue cells are complementary (positive), red cells substitutive/redundant (negative); the diagonal main effect is masked. Substitutive pairs are within-axis; cross-axis pairs, including C5×G1, are weakly complementary. Source: `outputs/shap_interactions.csv`.